



A walk-forward multi-model approach to stock trading: leveraging sentiment, trend indicators, and transformer-based deep learning for next-day price forecasting

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Abstract

Forecasting stock prices is complicate due the financial market's nature, which includes volatility, non-stationarity, and reliance on information for decision-making. Stock prices respond to a multitude of information sources including technical indicators, historical price movements, news articles, and investor sentiments. The recent introduction of FinBERT-based transformer models has also improved prediction accuracy; however, the majority of existing methodologies utilize static train-test splits, limited feature selection, weak cross-modal fusion, and do not take into account the impact of changing market regime dynamics, resulting in lower-than-expected robustness when applied to changing market conditions. This paper presents a new hybrid forecasting framework that enhances the prediction ability using walk-forward training; performs cross-modal feature selection; includes awareness of market regime; and reduces prediction bias with the use of a meta-classifier for the final prediction output. The framework developed in this study combines the use of sentiment features from FinBERT with price-based technical indicators to conduct stock price forecasts. Hidden Markov Models are used to identify market regimes, and the temporal dependencies is learned through the application of a CNN-Transformer architecture. Mutual information based feature selection is used to reduce the redundancy of features, and the final stock price forecast will be produced through the application of a meta-classifier. A walk-forward evaluation is used in order to ensure robustness to non-stationary stock price movement data. The average accuracy for experimental results on ten major U.S. stocks is 70%, F1-macro is 68.0%, MCC is 61.6% and AUC is 76.5%. Backtests incorporating realistic transaction costs produced a win rate greater than 60.07% and average returns of 30.82% which exceeded the performance of a pure deep learning-only baselines by a factor of ($p < 0.05$). Thus, this work presents a robust, regime-aware and leakage resistant stock forecasting framework which displays superior predictive and trading capabilities when compared to other methods simulating actual market conditions.

Keywords FinBERT · Technical indicators · CNN–transformer · Meta-classifier stacking · Walk-forward validation

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1 Introduction

Stock market prediction is an extremely difficult endeavor due to the complex, nonlinear dynamics of stock prices, even in the simplest cases where prices are efficient in the marketplace [1]. In the past, technical analysis and quantitative models have been the main forecasting methods used in the study of stock price movements [2]. Most recently, advances in electronic trading platforms, Machine Learning (ML), and Natural Language Processing (NLP) have enabled the incorporation of additional data, namely textual sentiment, into stock price prediction models [3]. Sentiment analysis transforms unstructured textual information into

structured measurements, thereby allowing models to leverage population sentiment, mood, and behavioral patterns to predict stock prices [4]. This is why there is growing need to find models that can combine structured financial information with unstructured news text to enhance predictive power.

Previous studies mostly used recurrent neural networks (RNNs) and long short-term memory (LSTM) architectures, which are based on recurrent mechanisms to model sequential data. Over the past few years, financial natural language processing has undergone remarkable development, driven by the integration of transformer-based models, such as Bidirectional Encoder Representations from Transformers (BERT) and RoBERTa, and their domain-specific variants. Transformers, with the help of the self-attention mechanism, can successfully learn to memorize temporal features of text, thereby increasing their ability to encode long-term dependencies [5]. Transformer-based models are therefore especially good at predicting time-series data and identifying slow sentiment change. Financial BERT (FinBERT) is a domain-specific adaptation of BERT that is even more accurate at extracting sentiment in financial domains, having been trained on corpora specific to the domain. However, there are still enduring limitations to most developed sentiment-based systems, as they struggle to align more closely with time-series indicators, to adequately account for the delayed effects of events, and to trace persistent relationships in market behavior over time.

Technical indicators are essential tools for trend analysis, providing analysts with information about short-term price dynamics and broader market development. They are often used independently or combined with hybridization methods and machine learning algorithms [6]. However, signals generated by technical indicators are usually slow to respond to sudden market shifts triggered by news events or external shocks. To overcome this drawback, more recent studies have focused on minimizing lag by combining technical indicators and sentiment signals. In this way, trend indicator-sentiment signal hybridization strategies have been developed to improve the accuracy of forecasts. Empirical evidence shows that the use of sentiment derived from news stories and social media, as well as historical price data, enhances the quality of the forecasts made [7].

Forecasting stock exchange price dynamics using hybrid and fusion techniques has recently become more prominent; however, most preliminary research studies suffer from major deficiencies in model training and testing. Many studies, including those that use FinBERT with Convolutional Neural Networks (CNNs) or Transformer models, have used a single (80:20) training-test split in the past. Although this is a standard practice in the traditional machine learning setting, its use for forecasting financial markets is unsound

because the markets under consideration are usually not stationary. Financial markets often change their statistical characteristics with changes in economic and policy conditions and investor sentiment. Therefore, using a single training regime on fixed data introduces temporal bias and renders both the training and test data time-dependent, making it impossible to adapt the model effectively to incoming data as it flows. The models that are first trained on pre-determined datasets are more likely to generate overstated predicted earnings and, consequently, overstated agreement between predicted and actual future stock prices and earnings [8].

Many existing studies fail to explicitly model market regimes, mainly due to constraints imposed by their training and assessment models. These approaches often assume that the market forces are relatively stable in the long run. However, financial markets experience transitions between discrete regimes, including bullish, bearish, and high-volatility regimes, each associated with a unique risk profile and different sensitivities to informational inputs. Since the predictive relevance of variables varies across these regimes, a fixed predictive model is unlikely to maintain the highest predictive quality as the market changes.

Hybrid multi-modal models that combine textual sentiment features and price-derived technical indicators have been shown to be more accurate at predicting price movements. However, more recent studies are mostly based on fixed cross-modal feature-selection models that do not explicitly model the interaction between sentiment and technical signals. This leads to the situation in which redundant or lowly informative features are systematically retained, while truly complementary information is poorly exploited. Besides, the use of constant, temporally independent weighting across all features neglects the temporal variability in feature relevance across varying market environments. Indicatively, sentiment features are more effective in times when the market is experiencing news-related volatility, whereas technical indicators are better predictors in stable and trending markets.

In many available studies, the final predictions are produced by the underlying classifier without regard for the credibility or temporal coherence of its results. Since financial markets are non-stationary, this type of single-stage decision-making is susceptible to systematic biases during regime changes. This is because the inclusion of a meta-classifier at the decision level allows the model to learn by incorporating information about historical out-of-sample prediction behavior and, consequently, recalibrate terminal decisions, leading to greater robustness and generalization in environments with changing market conditions.

The paper introduces a new methodological framework for predicting next-day stock price trends using a

multi-modal hybrid learning approach. The model combines sentiment features from FinBERT with representations of price-based technical indicators generated by a transformer model. It is time-consistent and explicitly mindful of market regimes, and it uses a walk-forward training and evaluation method. Moreover, it also uses decision-level meta-learning to improve the robustness in a non-stationary market. The main contributions of this study are the following:

- **Time-consistent walk-forward multi-modal learning framework:** A strict walk-forward training and testing procedure is used where the various parts of the predictive pipeline, feature selection, regime inference, representation learning, and decision fusion, are only updated using historical data at each time junction. In contrast to the fixed train-test partitions used in previous studies, this approach eliminates temporal leakage and enables an authentic evaluation of performance under conditions reflective of real-life trading.
- **Cross-modal mutual information-based feature selection:** To address the redundancy and noise inherent in high-dimensional sets of technical indicators, we introduce a feature selection model based on mutual information. This model is an explicit measure of the statistical dependence between sentiment characteristics derived from textual news information and technical indicators. The method improves robustness and generalization over uni-modal or heuristic feature selection methods by using a cross-modal selection strategy that retains only complementary indicators.
- **Explicit and decoupled market regime modeling:** An independent Hidden Markov Model trained on historical price data is used to infer market regimes. Regime information can be fed in as contextual input and not a predictive shortcut by decoupling regime discovery from the prediction task. This design will enable adaptive decision-making across bullish, bearish, and sideways markets.
- **Regime-aware CNN-Transformer representation learning:** A convolutional neural network-transformer architecture is used in this study to jointly identify both short-term technical features and long-term temporal correlations of the price information. The regime information is present at the decision-making level via regime-consistent conditioning, enabling the model to learn regime-specific feature interactions without mixing regime identification and representation learning.
- **Adaptive feature-wise attention across modalities:** A feature-wise attention system continuously modulates the relative attention to sentiment-based and price-based features as the process progresses, enabling the model to emphasize sentiment-based features during news-driven

market trends and technical features during trend-dominated ones. This dynamic reweighting can address the limitations of static feature weighting in non-stationary financial markets.

- **Decision-level meta-learning under walk-forward constraints:** Instead of training with feature or embedding level fusion, a meta-classifier based on Random Forests is trained on out-of-sample probability estimates generated across several walk-forward folds. This decision-level learning step identifies systematic biases and regime-specific reliability patterns in base-model predictions, thereby enhancing calibration and minimizing variance across rise, neutral, and fall cases.

The remainder of the paper is organized as follows. Section 2 reviews related work on sentiment-based prediction, regime-aware modeling, and multi-modal financial forecasting. Section 3 outlines the data sources, pre-processing steps, and methodological design of the proposed framework. Section 4 details the experimental setup and model configurations. Section 5 presents the empirical results and analysis, and Section 6 discusses them. Finally, Section 7 concludes the study and discusses potential directions for future research.

2 Related work

It is difficult to predict future stock prices due to the highly non-linear and volatile nature of financial markets; consequently, stock price prediction is regarded as one of the most challenging tasks in understanding market behavior [9]. Stock prices are influenced by several contributing factors, including (1) historical price data, (2) trading volume, (3) macroeconomic indicators, (4) geopolitical events, and (5) investor psychology, such as value perception, fear of loss, and overreaction [10]. Stock market price forecasting is therefore performed in an environment characterized by high uncertainty, randomness, and rapidly changing dynamics, rather than one that requires absolute certainty or precision. As a result, the primary objective of an effective forecasting model is to reduce uncertainty by employing reliable, objective, and repeatable methodologies that can support investment and portfolio management decisions. Accordingly, this section reviews a wide range of stock price forecasting techniques, including classical statistical models, machine learning approaches, sentiment-based methods, and deep learning architectures [11]. The section concludes by identifying limitations in existing studies, thereby motivating the present research.

Stock market prediction tasks are generally formulated as either regression or classification problems. Regression

approaches aim to forecast exact price values at specific time points, whereas classification approaches seek to determine the directional movement of the market [10]. The accuracy of such forecasts depends heavily on data preparation quality, feature representation, and overall model design. Forecasting methodologies may rely exclusively on numerical indicators, textual sentiment information, or a combination of both [11]. Early research predominantly employed statistical models to capture stock price dynamics from historical data [12]. Among these, autoregressive integrated moving average (ARIMA) models were widely used due to their effectiveness in modeling stationary and linear time series. However, ARIMA models struggle to capture abrupt and unpredictable market fluctuations, which are common in real-world financial markets. Generalized autoregressive conditional heteroskedasticity (GARCH) models are frequently applied to forecast future volatility based on historical volatility patterns, but they exhibit similar limitations when confronted with non-linear market behavior and unexpected events [13]. The Kalman filter provides a recursive framework for estimating time-varying systems; however, its performance is highly sensitive to model noise and underlying assumptions [14]. As a result, despite their interpretability and strong theoretical foundations, these classical methods often fail to deliver high predictive accuracy in volatile and high-dimensional market environments.

Machine learning techniques, including Support Vector Machines, Decision Trees, and ensemble approaches such as Random Forests, have enabled the development of more effective stock forecasting models compared to traditional statistical methods [15]. These techniques are capable of modeling complex non-linear relationships across different stocks and market regimes, and have demonstrated superior predictive performance relative to classical approaches [16]. Nevertheless, their effectiveness remains highly dependent on feature preprocessing, and they often lack sufficient adaptability across varying market conditions. Consequently, feature extraction and hyperparameter tuning continue to pose significant challenges in real-world stock forecasting applications.

The adoption of deep learning has substantially enhanced stock prediction performance by enabling models to capture time-varying and high-dimensional interactions among financial instruments [11]. Commonly used architectures include recurrent neural networks (RNNs), long short-term memory (LSTM) networks, gated recurrent units (GRUs), and convolutional neural networks (CNNs) [11]. More advanced architectures, such as Bidirectional LSTMs and attention-based networks, have further improved the modeling of long-term temporal dependencies and localized patterns [17]. Hybrid models that integrate traditional techniques with deep learning approaches have also been

proposed to balance predictive accuracy and model stability [11]. For example, Selvan et al. introduced a hybrid LSTM–RBFN model for stock price prediction [17]. Despite these advances, technical analysis remains a cornerstone of stock forecasting. Common indicators used to assess trends and momentum include simple and exponential moving averages, relative strength index (RSI), and moving average convergence divergence (MACD) [18]. Recent studies have focused on enhancing trend detection using deep learning; for instance, Wen et al. applied wavelet transformations to LSTM networks to identify trends across multiple timeframes [18]. However, trend-based systems often perform poorly during sideways or highly volatile market conditions and typically lack adaptability.

Investor sentiment has received increasing research attention due to the influence of psychological and behavioral factors on price movements [19]. Early sentiment analysis approaches relied on lexicon-based scoring combined with traditional classifiers. The introduction of transformer-based models, such as BERT [19], RoBERTa [20], and FinBERT [21], marked a significant advancement in financial text representation. Among these, FinBERT is uniquely fine-tuned for financial data and has become widely adopted for market sentiment analysis [21]. Recent studies have demonstrated that integrating sentiment features with technical indicators using deep learning improves directional prediction accuracy [22]. Similarly, Khan et al. reported enhanced robustness across varying market conditions by combining sentiment analysis with technical indicators [23]. Researchers have also explored incorporating additional data sources, such as news articles and social media platforms, to further improve predictive performance [24, 25].

Transformer-based models originally developed for natural language processing, including the Temporal Fusion Transformer (TFT) [26] and attention-recurrent networks [27], have been adapted for stock market forecasting. These models offer strong capabilities in modeling long-range dependencies and integrating heterogeneous inputs, thereby improving forecasts that combine numerical indicators with textual sentiment [28]. Fusion-based forecasting systems further highlight the benefits of this multimodal approach. For example, Shaban et al. developed a hybrid BiGRU–LSTM model that outperformed conventional deep learning approaches [29], while Salemi et al. enhanced artificial neural networks with technical indicators to improve prediction accuracy [30]. Other multimodal fusion models, such as DCIFNet and mWDN–LSTM, demonstrate practical advantages in real-world forecasting scenarios [18]. Optimization-based model fusion strategies have also been shown to yield further performance gains [31]. Despite these advancements, deep learning models often lack interpretability compared to linear regression or support vector

regression methods [32], are prone to overfitting [33], and require large labeled datasets. Sentiment analysis introduces additional challenges, including semantic ambiguity and sarcasm [34]. Moreover, trend-based methods tend to underperform in volatile markets, and transformer architectures impose substantial computational demands [26].

More recent research has continued to develop hybrid architectures and adaptive forecasting frameworks targeting nonstationary financial time series. One of these contributions is an exploratory analysis that looked at hybrid deep learning models which combine structurally complementary architectures to better address structural breaks and changing dynamics [35]. This research thus provides a methodological basis for developing regime-aware forecasting systems. A comparable approach developed a walk-forward validation framework while training deep representations of news in conjunction with macro-financial indicators [36]. Their findings highlight the importance of performing sequential evaluations and implementing robust multiple testing procedures when working with sentiment-integrated neural networks. A notable study proposed a hybrid CNN-Transformer architecture that combines convolutional feature extraction with attention mechanisms for enhanced multivariate time series forecasting and is consistent with multimodal financial forecasting frameworks [37]. Additional studies build on the existing literature by creating a hybrid quantitative trading paradigm using ensemble learning, multimodal fusion mechanisms, and transfer learning; this is done to improve flexibility in predicting targets and generalizing across heterogeneous market environments and to increase robustness to regime shifts [38].

Recent studies indicate that combining sentiment features extracted using FinBERT with price-based technical indicators modeled through transformer architectures can significantly enhance stock trend forecasting accuracy. However, although the FinBERT-TFT framework proposed in [39] demonstrates superior predictive performance, it relies on a single time-based train-test split. As the model is trained only once and remains fixed during evaluation, its robustness under evolving market conditions and structural regime shifts is not fully assessed.

2.1 Limitations of existing works

- **Static training and evaluation protocols:** Existing hybrid Price-Sentiment Models implement a single, monolithic split of data into a training and testing set. This method is not suitable to non-static, variable financial market data and produces performance estimates that can be overly optimistic and do not translate well into future market environments.

- **Ignoring the concept of regime awareness:** The current approaches have established that most models assume that the behavior of financial markets has not changed over time; however, almost no models have explicitly modeled regime-based dynamics that lead to disruptions to the expected output following regime shifts, thereby undermining confidence in results when transitioning from one regime to another.
- **Static methods for fusing technical indicators and sentiment features:** Technical indicators and sentiment features are commonly combined (fused) using fixed or heuristic methods. These methods do not include any explicit modeling of cross-modal dependencies and do not take into account the dynamically changing nature of the relevance of technical indicators and sentiment features over time. Thus, there is significant duplication and noise introduced when developing a representation from multiple features from various modalities.
- **Lack of decision-level reliability modeling:** Existing frameworks operate with little to no integration of decision-level meta-learning or to assess the robustness of predictions made. As such, they have proven to be less robust than they could be given the changing nature of financial instruments and their corresponding market regimes.

These limitations necessitate the construction of a new methodological framework that employs consistent time-based, regime-based, and walk-forward methodologies along with decision-based meta-learning.

3 Methodology

This study aims at developing a multi-modal hybrid forecasting framework to differentiate market regimes, selectively extract features, and is trained using a walk-forward algorithm to predict stock market behavior. The framework is based on sentiment analytics and traditional market signals and uses Transformer-based deep learning frameworks to capture complex temporal relationships. Moreover, it uses an adaptive decision-level fusion mechanism that integrates heterogeneous information streams and employs a walk-forward training regime. The architectural design will aim to capture the short-term effects of the news releases on market behavior, as well as the long-term trends and feedback processes. All the system's constituent modules are designed to focus on different types of stock price dynamics.

3.1 Stock selection

The massive.io API is used to collect financial articles from a number of publicly listed multinational companies that comprise a mixed capital investment portfolio, including Apple, Google, Tesla, Boeing, ExxonMobil, Walmart, Pfizer, Schlumberger, Sony, and Intel. The content provided all the articles from January 2014 to December 2024. For the same period, the daily closing market values for each listed public company mentioned above are obtained from the yfinance API. The selected stocks, including their ticker symbols and the relevant source of the daily price information, as well as the date when the daily price data became available, can be found in Table 1.

3.2 Sentiment analysis

Investor perceptions and collective emotion often drive short-term price changes that cannot be fully accounted for by historical data alone. Therefore, incorporating the Analysis of Market Sentiment into the Financial Prediction Model has become increasingly important. When combined with price-based indicators, Sentiment Indicators provide additional information that would not be available otherwise. The information it provides reflects the Mood and Expectations of Investors and can play an important role in influencing Trading behavior and Asset Prices.

BERT is a transformer-based model that is commonly used in NLP. Due to its effective implementation in a wide range of general-language tasks in different fields of NLP, the model is built on unstructured data. Nevertheless, a specialized model is needed for specific domain tasks, such as financial language processing. To overcome these weaknesses, a variant of financial language processing, FinBERT, was developed through domain adaptation. A large-scale corpus of financial documents was used to fine-tune FinBERT. In architecture, FinBERT maintains the original BERT model, which has 12 transformer encoder layers, each with a hidden size of 768 dimensions and 12 self-attention heads. FinBERT uses Next Sentence Prediction and Masked Language Modeling during the pre-training stage to learn deep contextual representations. Once pre-training

is complete, FinBERT can predict the sentiment of text instances by using the contextual embedding of the special token and passing it through a softmax layer to produce sentiment probabilities. Figure 1 shows the overall architecture of FinBERT, adapted from [40].

Let $D = \{d_1, d_2, \dots, d_n\}$ denote the collection of news documents associated with a given stock. Each document d_i is processed using FinBERT to obtain class probabilities through the softmax output layer:

$$P(C = c \mid d_i), \quad c \in \{0, 1, 2\} \quad (1)$$

where 0, 1, and 2 correspond to neutral, negative, and positive sentiment classes, respectively. The predicted sentiment label is determined as:

$$\text{Sentiment}(d_i) = \arg \max_{c \in \{0, 1, 2\}} P(C = c \mid d_i) \quad (2)$$

To incorporate prediction confidence, a sentiment confidence score α_i is defined as:

$$\alpha_i = \max_{c \in \{0, 1, 2\}} P(C = c \mid d_i) \quad (3)$$

The final sentiment representation for each document is expressed as a weighted sentiment vector:

$$\tilde{s}_i = \alpha_i \cdot \text{Sentiment}(d_i) \quad (4)$$

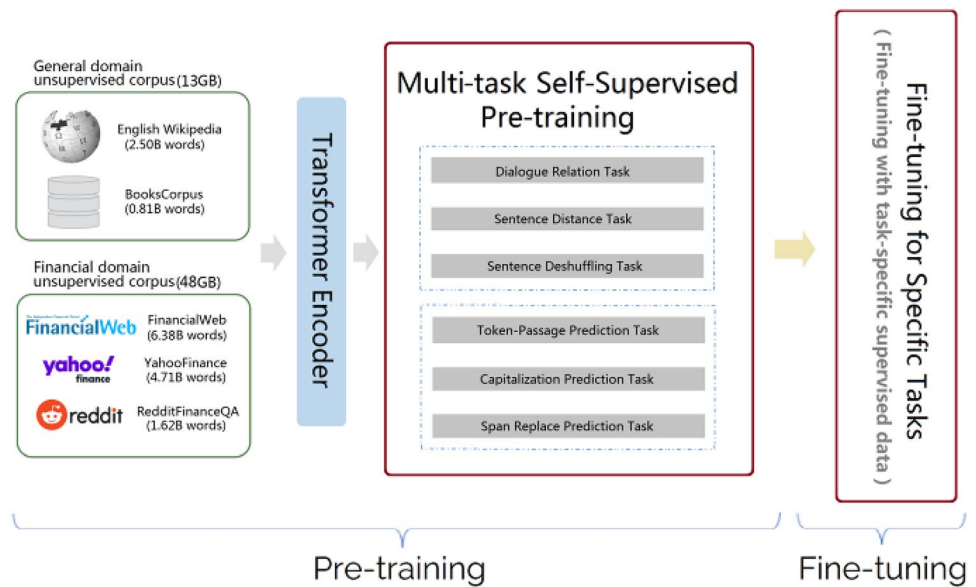
which reflects both the sentiment direction and the model's certainty. These weighted sentiment vectors serve as numerical features in the proposed hybrid forecasting framework. Rather than using a typical softmax layer to derive sentiment scores for positive, negative, and neutral sentiment classification, the proposed hybrid Forecasting framework uses a 768-dimensional dense embedding vector that captures more contextual and semantic detail than probability-based classification alone.

Dimensionality reduction via principal components analysis (PCA): The 768-dimensional FinBERT representations provide a wealth of information from a semantic

Table 1 Details of experimental datasets

Stock (ticker)	Financial news data	Stock price data source	Duration
Apple (AAPL)	massive.io API	yfinance	Jan 2014 – Dec 2024
Google (GOOGL)	massive.io API	yfinance	Jan 2014 – Dec 2024
Tesla (TSLA)	massive.io API	yfinance	Jan 2014 – Dec 2024
Boeing (BA)	massive.io API	yfinance	Jan 2014 – Dec 2024
Exxon Mobil (XOM)	massive.io API	yfinance	Jan 2014 – Dec 2024
Walmart (WMT)	massive.io API	yfinance	Jan 2014 – Dec 2024
Pfizer (PFE)	massive.io API	yfinance	Jan 2014 – Dec 2024
Schlumberger (SLB)	massive.io API	yfinance	Jan 2014 – Dec 2024
Sony (SONY)	massive.io API	yfinance	Jan 2014 – Dec 2024
Intel (INTC)	massive.io API	yfinance	Jan 2014 – Dec 2024

Fig. 1 FinBERT architecture adapted from [40]



perspective; however, since our training samples are so small compared to the feature dimension, using such a large feature dimension also significantly increases the amount of computation needed to do analysis and poses a risk of overfitting on these small training sets. To solve this problem, we employ PCA, which projects the representations onto a lower dimensional space that retains the most variance, and then constructs the embedding matrix $E \in \mathbb{R}^{T \times 768}$ (where T is the number of trading days), computes its covariance structure, and extracts the top $k = 8$ principal components, which together capture around 95% of the variance. Thus, the transformation we used to achieve this projection is:

$$\tilde{e}_t = W^T(e_t - \mu) \quad (5)$$

where W is a matrix (768×8) that is populated with the eight largest eigenvalues; μ is a vector for the mean embedding; and $\tilde{e}_t$ is a vector for a compressed daily sentiment feature. The eight dimensions for these vectors encapsulate the underlying structures of the main variances associated with financial news sentiments while significantly decreasing both the number of calculations and the chance of the model becoming overfit by noise in a high-dimensional environment.

3.3 Trend-following technique

Trend following is one of the primary principles in traditional stock market trading. When someone invests in a stock through traditional means, they would buy it at the moment they see that the stock has begun to rise and sell it when they see that the stock has begun to fall. The decision to buy or sell is primarily based on the price chart and various technical indicators. The approach in this paper used the

same philosophical approach to trend following, but instead of an investor determining when to buy or sell based on a trend, the model was trained on trend-following indicators that enabled it to learn from its experience to make its own decisions. Potential patterns were extracted for use in the model training process by applying technical indicators to retrieve interpretive and temporal signals. The technical indicators chosen for this work are well-known technical indicators that have been used for many years for the purpose of identifying trends, estimating future volatility, and forecasting. The closing price of stock P_t represents the closing price at time t . Technical indicators used for analysis include:

- **Simple moving average (SMA):** The price series is smoothed by taking averages over "n" periods; short-term fluctuations are reduced. Defined as:

$$SMA_t^{(n)} = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \quad (6)$$

- **Relative strength index (RSI):** The momentum was calculated by looking at the strength of the most current up and down movements (Gains minus Losses) over a look-back period defined as n :

$$RSI_t = 100 - \frac{100}{1 + RS_t}, \quad RS_t = \frac{\bar{G}_t}{\bar{L}_t} \quad (7)$$

where $\bar{G}_t$ and $\bar{L}_t$ denoted the average gains and losses over the last n periods.

- **Moving average convergence divergence (MACD):** The difference between a 12-period and 26-period Exponential Moving Average (EMA) highlights any changes in price trends:

$$\text{MACD}_t = \text{EMA}_t^{(12)} - \text{EMA}_t^{(26)} \quad (8)$$

A signal line (9-day EMA of MACD) was also calculated to detect crossovers.

- **Bollinger Bands (BB):** Volatility is captured with a measure of Standard Deviations (SD) above and below a Simple Moving Average, which is determined using:

$$\text{Upper}_t = \mu_t + k\sigma_t, \quad \text{Lower}_t = \mu_t - k\sigma_t \quad (9)$$

where μ_t was the n -period SMA and σ_t was the standard deviation of prices over the same window. Typically, $k = 2$. For every stock in the dataset, technical indicators, including the trend-following indicators, were used as additional predictors along with the Sequence-Based Learning and Sentiment Analysis components to improve the overall accuracy of predictions achieved by the model.

3.4 Transformer model

Transformers are one of the classes of Deep Learning models that are intended to deal with sequential data using attention-based methods [41]. In contrast to the traditional RNN and CNN approaches, which use a recurrence or a convolutional operation, respectively, Transformers process the complete input simultaneously and assign weights at

different points within the input sequence using the attention mechanism. This design allows transformers to capture both local and long-range correlations between inputs or events. In addition, since many time-series datasets, such as financial data, contain both regimes and lagging events, transformers have been especially promising.

Of the different transformer architectures, encoder-only models are the most appropriate for use in time-series situations due to their ability to model representations. The encoder portion of the transformer is built from a number of stacked layers, each of which contains two basic components: a multi-head self-attention (MSA) block and a position-wise feed-forward network (FFN) block, as shown in Figure 2. The MSA computes interactions between every single pair of tokens in the sequence and assigns importance weights to each token based on that interaction. This enables the transformer to represent richer contextual dependencies than previous models, which relied solely on either recursion or convolutions to extract contextual information. The MSA mechanism is formally defined as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (10)$$

where Q , K , and V represent a Query, Key, and Value vector, respectively, that are derived from the input embeddings, where d_k is equal to the dimension of the key space. To improve the ability to learn representations of data, the transformer uses Multi-Head Attention with each of the heads computing attention over a different subspace of the input. This approach increases the models' ability to learn about complementary patterns and relationships between the elements within the input sequence.

The computation of Multi-Head Attention can be expressed mathematically in the following equation:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O, \quad (11)$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad (12)$$

where W_i^Q , W_i^K , W_i^V , and W^O were learnable projection matrices.

Figure 2 also highlights the components of the encoder-based Transformer architecture. The encoder-only Transformer performed particularly well on our financial time series data. Due to the encoder's ability to use parallel processing, the encoder-based model could use a greater amount of data that combined textual sentiment data, technical analysis, and fundamental economic data inputs [28]. By allowing us to use attention weights, we could visualize and interpret which events or features were most important when it came to making the predictions. This type of

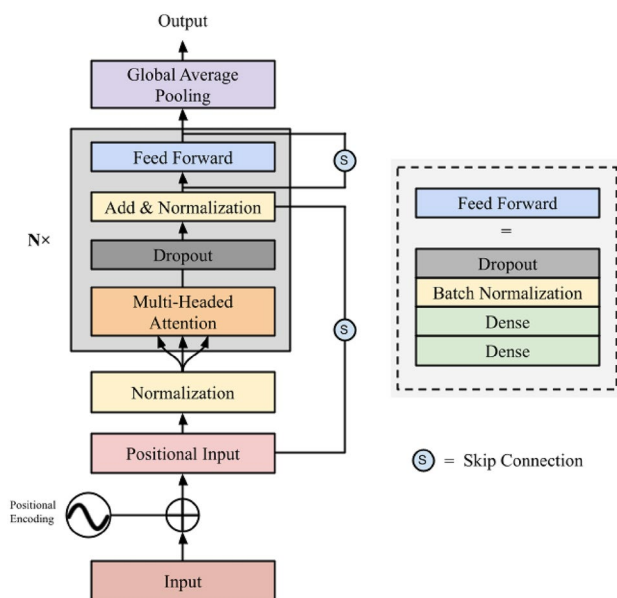


Fig. 2 Encoder-only transformer architecture [42]



Fig. 3 High level illustration of walk forward training [46]

visualization can help investors with greater insight when developing trading strategies [32]. In combination with a hybrid approach combining instantaneous sentiment extraction via FinBERT and time series trend indicators, the encoder-only Transformer improved both prediction performance and resilience in uncertain markets [29, 30].

3.5 Convolution neural network, walk-forward strategy, and meta-classifier

A CNN is the principal feature extractor in this proposed hybrid forecasting framework, which uses 1D convolutional filters to process financial time series data. CNNs excel at identifying local spatial patterns that exist within sequential data and can provide an indication of future market behaviour. A typical CNN has several filter sizes and ReLU activation layers, and may have pooling layers to reduce noise and preserve key features from each filter. Once features have been extracted through the CNN, the features are passed to the transformer encoder, which models the long-term time dependence and the surrounding context in the financial market. [43–45].

A time series model can be evaluated realistically and unbiasedly by using Walk-Forward training. The model is trained in an iterative manner using the data up to that point, and the model is tested on the next time period, preserving temporal causation strictly. The importance of walk-forward validation is significant in financial time series forecasting due to the inherent non-stationarity of financial market data, due to the volatility and structural change associated with market data, as well as the changes in the relationships between the series over time. Therefore, in contrast to random train-test splits that can create the potential for information leakage from future observations, walk forward validation provides a way to avoid creating the potential for leakage into the future while ensuring that all forecasts will be based only on what would have been known at the time of the forecast. [35]. The model is initially trained on

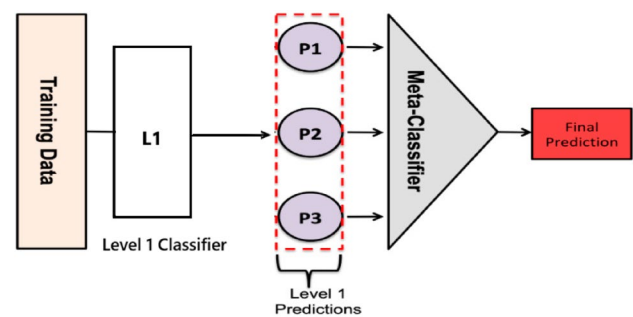


Fig. 4 Diagram of meta classifier

a set of data that is only available at the time of training. Once the model has been developed, it is then retrained as additional, more current data is added to the training set. The model continues to be retrained until all the available data has been used to validate the model, as illustrated in Figure 3. This means that every time the training window moves to the future, there is new data that has been used to validate the model. Therefore, the model will continue to evolve based on what has happened in the past, while still providing enough historical information for validation. Additionally, the training framework can provide a high level of robustness for predictions across a variety of market conditions, enabling the model to quickly adapt to changing market trends.

In addition, the training framework provides numerous opportunities for temporally consistent input to the meta-classifier window.

Base classifier, like a hybrid CNN-Transformer model in our case, might be biased in the case of non-stationary time series data. To address this problem and make the model robust, the Meta Classifier is helpful [47]. It selects optimal predictions from the base classifier to enhance performance. In this work, the Meta-classifier is essentially a Random Forest ensemble that derives its predictions based on the probability distribution of the CNN-Transformer outputs [37]. Figure 4 shows a high-level diagram of the meta classifier. The Random Forest uses these probabilities to determine patterns that the CNN-Transformer may not have captured on its own [48]. For example, a Random Forest may correctly predict that an asset will perform well during bull markets while not being able to identify the pattern in markets that are range-bound, through its probabilistic assessment of the CNN-Transformer predictions. Thus, the aggregate approach of Random Forest builds on the individual trees, which means that it reduces noise and overfitting, while also identifying and capturing complex non-linear relationships in the feature space [49].

Combining CNN and Transformer technology in conjunction with a Random Forest creates a comprehensive forecasting pipeline that is stronger than any of the individual

components on their own. The CNN identifies short-term patterns, the Transformer handles long-term dependencies, and the walk-forward evaluation ensures that the integrity of the data and the ability to respond to changes in the market through the use of an adaptable forecasting framework remain intact. The Random Forest takes the predictive outputs from the other two components, identifies the best prediction, and provides a more stable and accurate final prediction across the different types of markets throughout the changing market conditions [50].

4 Proposed framework

Forecasting whether the price of a stock will increase or decrease over the next day is challenging due to the simultaneous influence of numerous factors. These include historical price dynamics, technical indicators, and external news sentiment. Although technical indicators can provide insight into market behaviours, they tend to work best with clear trends and to be less effective in sideways markets or unexpected regime changes. While incorporating sentiment and news signals provides important context for stocks, their level of effectiveness is also dependent upon the current state of the market. Previously developed multimodal approaches for forecasting stock prices largely address this challenge through static feature fusion or end-to-end deep architectures. Such models implicitly assume the stock market as stationary and ignore temporal dependency and market regime.

In order to overcome the shortcomings identified above, a novel multimodal hybrid forecasting framework has been developed that is regime-aware and preserves temporal relationships across different data modalities. The framework explicitly models (i) cross-modal information dependency, (ii) market regimes, and (iii) temporal generalization under walk-forward constraints. The proposed model consists of four tightly coupled components:

1. **Cross-modal feature selection via mutual information:** We determine the statistical correlation of all News Sentiment features with Technical Indicators using the mutual information between them rather than picking Technical Indicators separately. By doing so, we can keep only those Technical Indicators that offer additional, non-redundant data about the corresponding News Sentiment feature. This helps to eliminate duplicate data and reduces the amount of "noise" in the dataset.
2. **Explicit regime discovery using an independent Hidden Markov Model (HMM):** Utilizing past price

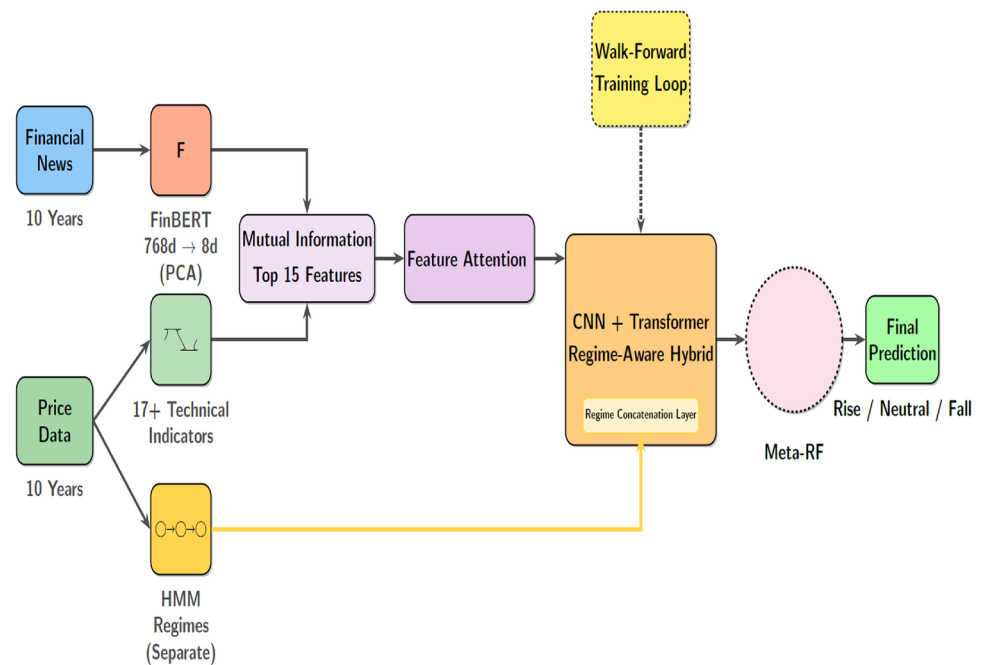
history as input into a HMM, the HMM identifies market nature (Up; Down; Neutral) independently of forecasts which avoids forecast leakage of historical price behavior and allows traders to use the regime classification as background comprehension.

3. **Regime-aware CNN–Transformer representation learning:** A CNN–Transformer encoder captures both short-term technical patterns and long-range temporal dependencies. A regime-consistent conditioning function is used at the decision layer to inject regime information directly into the model, allowing the model to learn how to capture regime-specific feature interactions.
4. **Walk-forward decision-level fusion via a meta-classifier:** This proposed hybrid forecasting framework executes decision level fusion in place of feature or embedding level fusion, while doing this in an out-of-sample probabilistic stream resulting from strict walk-forward validation. The decision level fusion enables the random forest meta classifier to learn the systematic biases of the base predictor and other regime dependent reliability through temporal dependency.

Figure 5 illustrates the overall proposed hybrid forecasting framework comprising two parallel information streams. The first stream processes financial news sentiment using FinBERT, producing 768-dimensional embeddings that are compressed to eight principal components via PCA for noise reduction and computational efficiency. The second stream extracts more than seventy technical indicators listed in Table 2 from daily price data. These two streams are merged into a unified daily feature vector. Mutual information analysis is then applied across modalities to retain the top 15 indicators that exhibit the strongest dependency with sentiment features.

A feature-wise attention layer is applied to dynamically reweight individual features at each time step, allowing the model to emphasize sentiment-dominant or price-dominant signals depending on market context. The attended features are passed through two convolutional blocks, designed to capture short-term patterns such as momentum bursts or volume surges, followed by two Transformer blocks that model longer-term temporal dependencies and recurring price structures.

In parallel, the HMM operates independently on historical price sequences to infer the prevailing market regime. The inferred regime is encoded using a one-hot representation and concatenated with the final dense representation of the CNN–Transformer encoder immediately before prediction. This design ensures that pattern

Fig. 5 Proposed novel hybrid forecasting framework**Table 2** Technical and auxiliary features used in the proposed hybrid forecasting framework

Category	Feature name	Description / parameters
Trend (10)	Returns	Daily price returns (% change)
	SMA_10	Simple Moving Average (10-day)
	SMA_30	Simple Moving Average (30-day)
	SMA_60	Simple Moving Average (60-day)
	EMA_10	Exponential Moving Average (10-day)
	EMA_30	Exponential Moving Average (30-day)
	Price_to_SMA30	Price relative to 30-day SMA (%)
	MACD	Moving Average Convergence Divergence
	MACD_signal	Signal line of MACD
	MACD_histogram	Difference between MACD and signal line
Momentum (5)	RSI_14	Relative Strength Index (14-period)
	Stoch_K	Stochastic Oscillator %K
	ADX	Average Directional Index (trend strength)
	ROC_10	Rate of Change (10-day)
	Williams_R	Williams %R momentum indicator
Volatility (3)	Volatility_20	20-day rolling standard deviation of returns
	BB_width	Normalized Bollinger Bands width
	ATR_pct	Average True Range (% of price)
Volume (2)	OBV_pct_change	Percentage change in On-Balance Volume
	MFI	Money Flow Index

learning remains regime-aware, while the final decision is explicitly conditioned on the current market state.

The entire training and evaluation process is embedded within a strict walk-forward validation framework, as shown in Figure 6. All components—including mutual information-based feature selection, regime inference, representation learning, and prediction—are trained sequentially using only past data, thereby eliminating look-ahead bias and ensuring temporal dependency.

Across ten years of data and 31 walk-forward folds, the base CNN–Transformer model produces out-of-sample class probabilities for each trading day. These probability streams, together with the corresponding realized outcomes, are used to train a random forest meta-classifier. The meta-classifier functions as a temporal decision aggregator, learning systematic biases and regime-dependent reliability patterns in the base model’s predictions. By operating exclusively on forward predictions,

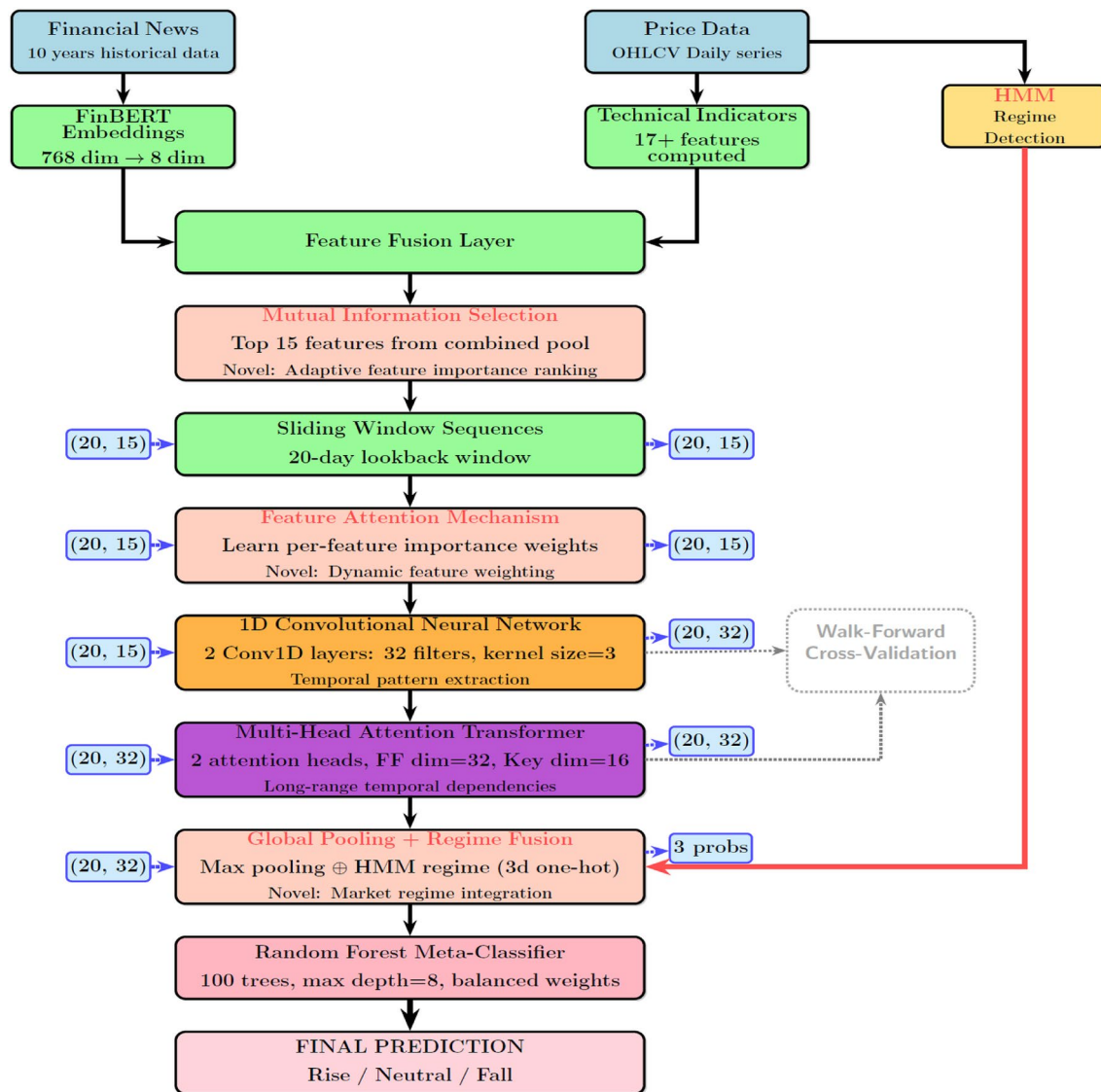


Fig. 6 System-level workflow of the proposed framework

this secondary learning stage improves calibration and robustness without introducing data leakage.

Overall, the proposed framework differs fundamentally from existing hybrid sentiment–technical models by combining cross-modal information-theoretic feature selection, explicit regime conditioning, and walk-forward decision-level fusion. This results in a realistic and deployment-ready prediction system that aligns model learning with the non-stationary and regime-dependent nature of financial markets.

4.1 Sentiment embedding extraction with FinBERT

Financial news articles are filtered according to market trading hours prior to conducting sentiment analysis. Therefore,

any financial news that is published after the U.S. stock market closes will not be processed until the next trading day since it did not have an opportunity to affect the market during that trading day. By processing the news this way, the timing between when the sentiment signal is produced and when the market reacts in relation to it remains intact. In the event that there are multiple pieces of financial news on a given day, all of the various sentiment signals will be combined together and then the result will be evaluated at the end of the trading day. If no financial news has been published on a given trading day, then the sentiment vector for the previous day will be carried forward into that trading day to allow for continuity in the time series associated with the sentiment vector. This process is illustrated in Algorithm 1.

```

1: for each trading day  $t$  in the dataset do
2:   Collect all news articles published before the market close of day  $t$ 
3:   if news articles are available then
4:     Compute FinBERT embeddings for the collected articles
5:   else
6:     Forward-fill the embedding from the previous trading day
7:   end if
8:   Use the embedding at day  $t$  to forecast the price movement between day  $t$  and
   day  $t + 1$ 
9: end for

```

Algorithm 1 Temporal alignment of news sentiment for price forecasting

This study utilizes FinBERT not to classify sentiment but instead to extract features from text. Instead of relying on the last softmax layer of the FinBERT model to create discrete sentiment classes from input data, it extracted the dense, contextualized embeddings directly from the encoder component of the model, which contain a lot of semantic and sentiment information.

Each financial news article d is tokenized into L subword tokens and passed through the FinBERT encoder, producing contextualized token embeddings $H(d) \in \mathbb{R}^{L \times h}$, where L denotes the number of tokens and $h = 768$ is the hidden embedding dimension. To obtain a fixed-length representation of the article, the token embeddings are aggregated using masked mean pooling, where padding tokens are excluded using an attention mask m . Specifically, the article-level embedding $E(d)$ is computed by summing the masked token embeddings and normalizing by the total number of valid tokens:

$$E(d) = \frac{\sum_{t=1}^L m_t \cdot H(d)_{t,:}}{\sum_{t=1}^L m_t + \varepsilon}, \quad (13)$$

where ε is a small constant added to prevent division-by-zero errors.

For each trading day t , the embeddings of all news articles published on that day are averaged to compute the aggregated daily sentiment representation $\bar{E}_t$. Let m_t denote the number of articles released on day t ; the daily sentiment vector is then given by

$$\bar{E}_t = \frac{1}{m_t} \sum_{j=1}^{m_t} E(d_{t,j}). \quad (14)$$

Finally, Principal Component Analysis (PCA) is applied to the daily sentiment vectors $\bar{E}_t$ to project them into a lower-dimensional space, reducing redundancy and overfitting while preserving the most informative variance in the sentiment representations.

4.2 Mutual information-based features selection for sentiment and technical indicators

To create a compiled feature vector for each day of trading, we combined the sentiment embeddings and technical indicators into one feature vector. From an initial pool of 20 technical indicators and 8 sentiment features, mutual information-based feature selection is employed to identify the top 15 features that provide the highest predictive relevance with respect to the target labels while minimizing redundancy. This approach addresses a fundamental challenge in quantitative finance, where many technical indicators are highly correlated. Mutual Information (MI) quantifies the amount of information obtained about one random variable by observing another, thereby measuring statistical dependence without assuming linearity. For a given feature X_i and target label Y , MI represents the reduction in uncertainty about Y when X_i is known. Within each fold of the walk-forward (WF) training scheme, the mutual information between a feature X_i and the target variable Y is computed exclusively using the training data. The mutual information is mathematically defined as

$$I(X_i; Y) = \sum_{x \in X_i} \sum_{y \in Y} p_{\text{train}}(x, y) \log \left(\frac{p_{\text{train}}(x, y)}{p_{\text{train}}(x) p_{\text{train}}(y)} \right), \quad (15)$$

where $p_{\text{train}}(x, y)$ denotes the joint probability distribution of X_i and Y , and $p_{\text{train}}(x)$ and $p_{\text{train}}(y)$ represent their respective marginal distributions, all estimated solely from the training data. The mutual information score is measured in bits or nats; a higher MI score indicates a stronger statistical dependency between the feature and the target variable. Conversely, a lower MI score suggests that the feature provides limited unique information, either due to weak relevance or significant overlap with other already selected features. The feature selection procedure for each walk-forward fold proceeds as follows:

1. Compute the mutual information score $I(X_i; Y)$ for all features using only the training subset of the fold.

2. Rank the features in descending order based on their MI scores.
3. Select the top 15 features with the highest mutual information values.
4. Use the selected features consistently for both training and validation data within the corresponding fold.

While feature selection based on MI ranks features independently, MI has the ability to detect both linear and nonlinear relationships between the features and the target variable. By using convolutional layers, nonlinear activations, attentional mechanisms, and a Random Forest meta-classifier, the CNN/Transformer model learns about Nonlinear dependencies between the selected features and their interactions. In this manner, any Nonlinear feature dependencies are dealt with properly during all of the learning phases of the framework.

4.3 Attention-based features weights

Multimodal financial prediction is inherently challenging, as each of the input features contributes differently under varying market conditions. For instance, during earnings announcements, sentiment extracted from financial news is expected to play a more significant role, whereas during technical breakouts, price-based technical indicators become more influential. Treating all features with equal importance, therefore, limits the model's ability to adapt to such changing dynamics. Instead of uniformly averaging the inputs, the proposed Attention-Based Features Weights mechanism enables adaptive computation of feature weights by learning their predictive relevance at each time step. Let the input feature matrix be denoted as

$$X_0 \in \mathbb{R}^{T \times d} \quad (16)$$

where $T = 20$ represents the number of time steps and d denotes the number of input features. The feature-wise attention scores are computed as

$$A = \text{softmax} \left(W_2 \cdot \text{ReLU} \left(W_1 X_0^\top \right) \right), \quad (17)$$

where W_1 is a weight matrix that projects feature vectors of dimension d into a hidden representation of size h (with $h = 16$), followed by a ReLU activation function to introduce non-linearity. The matrix $W_2 \in \mathbb{R}^{d \times h}$ then maps the hidden representation back into the original feature space. The softmax function normalizes the attention weights such that they sum to one across all feature dimensions for each time step, producing normalized feature importance scores.

The attended feature representation at time step t is obtained as

$$X'_t = X_t \odot A, \quad (18)$$

where $\odot$ denotes element-wise product. This operation amplifies informative features while suppressing noisy or less relevant ones.

The use of two learnable projection matrices, W_1 and W_2 , allows the attention mechanism to capture non-linear relationships among features rather than restricting the model to linear combinations. During training, gradients propagated from the prediction loss enable the attention module to learn context-dependent feature relevance. For example, under high-volatility conditions, higher attention weights may be assigned to indicators such as Average True Range and Bollinger Band Width, whereas in trending markets, momentum-based indicators such as MACD may receive greater emphasis. This adaptive weighting strategy not only improves predictive accuracy but also enhances model interpretability, as the learned attention weights directly reflect the features that contributed most to each prediction.

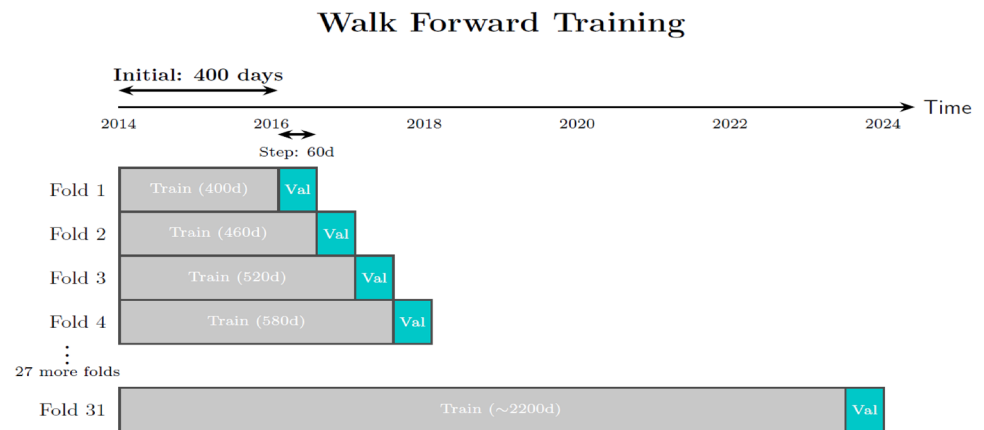
4.4 CNN–transformer encoder

As highlighted in Fig 6 the CNN-Transformer encoder receives sequences containing L days in length, where every day is represented by the feature vector created when the shortened sentiment embedding is combined with the computed set of technical indicators associated to those same days. This allows the model to simultaneously evaluate both textual sentiment trends and numerical indicators related to the stock market.

A 1-D CNN captures short-term dependencies and local temporal patterns over adjacent days. To preserve temporal ordering prior to passing the CNN output into the self-attention module, the CNN output has a positional embedding for the days that is learnable. Once passed into the multi-head self-attention layer, the embeddings for each of the L days are combined such that they model long-range relationships between the L days and thus provide the model with a full understanding of the overall relationship between textual sentiment trends and numerical indicators related to the underlying market. In addition, during this layer, a multi-layer feed-forward network provides non-linear transformation to the previous set of output from the first layer (CNN) and the last layer (multi-head self-attention). Finally, all of the previous layer outputs are pooled into a global representation corresponding to each input sequence, which will be used for classification.

4.5 Walk-forward training and meta-classification

The framework proposed will realistically simulate the process of executing a trade by adopting a walk-forward

Fig. 7 Walk-forward training illustration

training strategy, as depicted in Figure 7. The first step of this proposed process is to use 400 historical trading days to build the first model, which will serve as the "training period" of the model. After training, the model will be evaluated over a 60-day period with no update of the training model. Each cycle will be used to produce out-of-sample predictions, with the 60-day testing evaluation added to the training data, while the evaluation period is likewise shifted forward in time.

The 400-day training period was intended to identify seasonal and medium-term shifts in market trends while also being quick enough to react to regime changes. Therefore, it allows enough historic information required to develop a stable model without using too much outdated information. The 60-day testing period allows a reasonable out-of-sample evaluation period and better models the way that markets actually operate.

Evaluation of possible configurations for training time with 300 days, 400 days and 500 days along with validation period of 50 days, 70 days and 80 days determined that a combination of 400 days training and 60 days validation/testing gives the best trade-off between model stability and responsiveness. Therefore, all of the results in this paper are from the 400-day training period and the 60-day validation period.

With daily historical data for nearly 10 years, this arrangement produces 31 "walk-forward" folds. Predictions made during the "out-of-sample" portions of the walk-forward period by the CNN-Transformer will be stored and will later be combined in order to produce the complete prediction of market conditions via the Meta-Classifer. The proposed method prevents leakage of historical data as well as preserves the strict time sequence of the data. The specific characteristics of the approach are that:

1. The CNN-Transformer encoder is trained exclusively on past data.

2. Out-of-sample predictions (class probabilities) are generated for the subsequent unseen period.
3. Predictions are stored, and the training window is advanced forward in time.

After an aggregate probability sequence for each prediction is collected, this aggregate will serve to train a Random Forest meta-classifier, which uses the probabilities of the predictions to identify non-linear decision boundaries in the probability space. In this manner, the systematic bias produced by using the base model can be reduced, as well as creating a robust method for responding to changes in a Market Regime.

Because of the mixed nature of this architecture, it enables deep learning to account for time-based relationships when analyzing data via time series analysis, using technical indicators, such as moving averages, to measure structured market dynamics, and using sentiment embeddings as exogenous variables when predicting price movements in financial markets. This creates a more effective and robust predictive framework.

5 Experimentation

The experimental results of the proposed novel hybrid forecasting framework on experimental data are two-fold. First, the proposed framework used for next day ($t + 1$) close price trend forecasting compared to the previous day (t) close price. Second backtesting result of the proposed framework. The experimental design for evaluating the proposed hybrid forecasting framework is presented in this section. It outlines the system configuration, data pre-processing steps, label construction logic, backtesting strategy, and overall experimental procedure. All model building, data processing, and evaluation were done in Google Colab, allowing for easy reproducibility and efficient resources. We

used the same random seeds across all libraries, leading to reproducible results.

5.1 Data preprocessing

Financial news headlines and summaries for ten different companies were gathered through the `massive.io` API for the construction of a consistent dataset that can be used in a hybrid forecasting model. The dataset as shown in Table 1 used for the experiment. Standardization of the text data occurred through the conversion of all text to lower-case letters, the removal of all punctuation marks and other non-ASCII characters, as well as the elimination of all tokens irrelevant to the topic. Following text data standardization, processed headlines were tokenized based on the FinBERT tokenization scheme and a batch generation of sentiment scores was performed. Subsequently, for each batch sentiment score generation, each individual day's sentiment score was averaged to determine the total value of the day's combined sentiment score. To enhance the reliability of sentiment score data, sentiment score weighting was determined by the model's level of confidence. Finally, through this process of generating daily aggregated sentiment scores consistent with stock trading sessions, multiple news stories published on the same day can be grouped together into one news-based entity and assigned one aggregated sentiment score.

Similarly, daily stock price information for the same companies and period were sourced using the `yfinance` API. The daily dataset was modified to include only valid trading days, so interpolation methods were used to fill any missing values. Then, sequential datasets were created that used a rolling window of the past 20 trading days. Therefore, each rolling window contained sentiment features and technical indicators and labeled with the price movement of the next trading day.

Temporal sequence construction: To create input sequences, a sliding window is used by taking a sample of 20 trading days (essentially 1 month). To create these input tensors, there is overlap in the input sequences, depending on the position of the current window within the available historical data.

By taking an input tensor based on a prediction made at day t , the model has access to previous historical data points for days $[t - 19, t - 18, \dots, t - 1, t]$, which results in input tensors shaped $(20, d_{\text{features}})$ where d_{features} represents the total number of dimensions across all features.

News temporal alignment: To achieve accurate forecasting and protect against possible information leaks, our proposed procedure will consider only news articles published prior to day t 's market close to forecast price movements between day t and day $t + 1$. In the absence of news

coverage, we will carry forward (or "forward fill") our FinBERT embedding results. As such, our method preserves the true temporal dependency between the availability of information and the price in a real-world trading environment.

Feature normalization: Z-score standardization is applied to all numerical variables so that they each have a mean of zero and a variance of one, when considering their respective dimensions. All features have been standardized using the `StandardScaler` method based solely on their scaling parameters were calculated on the training set of each walk-forward fold. The mean μ_{train} and the standard deviation σ_{train} computed from the training set were used for transforming the validation and test sets:

$$\tilde{x}_{i,t} = \frac{x_{i,t} - \mu_{\text{train},i}}{\sigma_{\text{train},i}} \quad (19)$$

This strict separation prevents data leakage from future periods and mirrors the information constraints of actual deployment.

5.2 Label construction and class definition

The percentage of the daily return on stocks is determined by the closings price changes. To be precise, daily returns at time t , which refer to as t_r , are calculated based on the closing price of stock for two consecutive days, which is reflected in Eq. 20 below. By using this method, you can compare the relative price movement of different assets regardless of their price.

$$r_t = \frac{C_t - C_{t-1}}{C_{t-1}} \times 100 \quad (20)$$

where C_t = the stock's closing on day t ; and C_{t-1} = the stock's closing on the prior trading day.

Using the calculated daily return r_t , class labels are assigned based on the rule defined in Equation 21. An asymmetric threshold of $\pm 0.8\%$ is applied to eliminate insignificant fluctuations in price due to market microstructure noise from the previous day and are unlikely to be accompanied by significant profit opportunities through trading. The selected $\pm 0.8\%$ value represents what would be considered typical transaction costs and volatility in the market studied; therefore, any fluctuations that are smaller than the $\pm 0.8\%$ threshold will generally be within the normal range of noise that occurs at any given point in time and will result in little or no profit when accounted for by trading cost and slippage. As such, only statistically and economically significant price shifts will create directional signals to improve the prediction model's overall robustness.

Fig. 8 Percentage distribution of Tesla (TSLA) daily price movement classes obtained using the labeling rule defined in Eq. (2)



Table 3 Transaction cost structure

Cost component	Basis points	Percentage
Commission & brokerage fees	10 bps	0.10%
Slippage (market impact)	5 bps	0.05%
Total per roundtrip	15 bps	0.15%

$$y_t = \begin{cases} 1, & \text{if } r_t \geq +0.8\% \quad (\text{Rise}) \\ 0, & \text{if } -0.8\% < r_t < +0.8\% \quad (\text{Neutral}) \\ 2, & \text{if } r_t \leq -0.8\% \quad (\text{Fall}) \end{cases} \quad (21)$$

where y_t denotes the class label of the stock on day t :

The design of a threshold has two distinct objectives. First, it addresses problems caused by label noise, created by random fluctuations in equity prices around zero. Second, it retains enough samples in both Rise and Fall classes to allow for effective learning.

When looking at the labeling rule applied to all ten stocks analyzed in this study, it is evident that the application has produced a reasonable and asset dependent class imbalance. The more stable the company, the greater percentage of days that will be included in the neutral class. Conversely, those stocks classified as being more volatile have a greater percentage of days included in the directional classes. For example, in Figure 8 is the class distribution for TESLA(TSLA), which is one of the most volatile stocks within this study, with a total of approximately 28% of trading days being classified as *Neutral* and the remaining

Table 4 Backtesting strategy variants

Strategy	Description
Standard	Trades on every model prediction regardless of confidence level. Higher trading frequency but includes uncertain signals.
Confidence-Based (primary approach used in this work)	Filters predictions below 65% maximum class probability. Trades only on high-conviction signals, reducing frequency and transaction costs while avoiding uncertain predictions. This is our primary approach.

observations being split roughly evenly between the *Rise* and *Fall* classifications of this particular stock.

5.3 Backtesting

For the purpose of converting the model's predicted results into a real-world trade, we emulate the way traders actually trade when using a model prediction. Our backtesting engine makes several assumptions about how it will operate, as shown in the tables below:

Key assumptions:

- Initial capital: \$100,000
- Position types: Long, Neutral, or Short
- Trading signals: Class 1 (Rise) → Buy, Class 2 (Fall) → Sell, Class 0 (Neutral) → Hold
- Position sizing: Full capital allocation per trade

Table 5 Example trading decisions with confidence filtering

Day	P (Rise)	P (Neutral)	P (Fall)	Max prob	Prediction	Action
1	0.72	0.18	0.10	0.72	Rise	Buy
2	0.45	0.35	0.20	0.45	Rise	Hold (low confidence)
3	0.15	0.25	0.60	0.60	Fall	Hold (low confidence)
4	0.12	0.15	0.73	0.73	Fall	Sell
5	0.55	0.30	0.15	0.55	Rise	Hold (low confidence)

- Market execution: Trade executed at next day's opening price
- No leverage constraints or margin requirements

The transaction cost structure that was used for each of the trades in the backtesting process is described in Table 3. Then, in Table 4, we summarise the two ways we have back-tested all the models.

Example: confidence-based trading decision: Table 5 illustrates a confidence-based trading decision mechanism using a five-day trading window to demonstrate how confidence filtering governs transaction execution.

On **Day 1**, the model predicts a price increase with a probability of 72%, which exceeds the predefined confidence threshold of 0.65. This triggers a *buy* signal. Consequently, shares are purchased at a market price of \$100.00 using the total available capital of \$100,000. Accounting for transaction costs, which include a commission fee of 0.10% and slippage of 0.05% (total cost of 0.15%), the effective number of shares acquired is calculated as

$$\text{Shares Purchased} = \frac{100,000}{100.00 \times (1 + 0.0015)} = 998.5 \text{ shares.} \quad (22)$$

On **Day 2** and **Day 3**, although the model predicts the market direction, the associated confidence levels remain below the threshold, with maximum probabilities of 45% and 60%, respectively. Due to insufficient confidence reinforcement, no new trades are executed, and the existing position is held to avoid premature or low-confidence transactions.

On **Day 4**, the model signals a probability of 73% for a price decline, surpassing the confidence threshold and triggering a *sell* signal. The shares are sold at a price of \$102.50. After incorporating transaction costs, the realized capital is computed as

$$\text{Capital Realized} = 998.5 \times 102.50 \times (1 - 0.0015) = \$102,169. \quad (23)$$

This trade yields a total net return of 2.17%, resulting in a realized profit of \$2,169 after accounting for all associated costs.

On **Day 5**, the model again reports a low confidence level, with the probability of price movement limited to 55%. Given the absence of a high-confidence signal, no trade is executed, and capital is retained in cash while awaiting a more reliable opportunity.

Overall, trades are executed on only two out of five trading days (40% participation). This selective execution strategy ensures that capital is allocated exclusively to high-confidence scenarios, thereby reducing exposure to uncertainty and minimizing the risk associated with erroneous or low-confidence trades.

5.4 Implementation details

5.4.1 System configuration

We used Google Colab Pro with a cloud-based GPU for our experiments. System configuration:

- **Hardware:** Nvidia Tesla T4 GPU (16GB VRAM), 25GB of RAM, Google Colab environment.
- **Software:** Python 3.10, TensorFlow 2.13, Keras, PyTorch 2.1, Scikit-Learn, Pandas, NumPy, Matplotlib.
- **APIs used:** YFinance for historical data from the stock market, Massive.io for news headlines and summaries.
- **Models used:** FinBERT from HuggingFace and the custom transformer model created in Keras.

5.4.2 Model architecture and training configuration

To ensure reproducibility of this work, we detail the architecture, training strategy, and hyperparameters used in each experiment. A summary of the configurations used is presented in Table 6 and 7, which outlines the model architectures. This includes the use of CNN for the front end, transformer encoding, and ensemble learning methods via Random Forest. The table also includes key information on optimizers, learning rates, CNN layer types, regularization, and walk forward validation so that others can precisely reproduce the experimental settings used in our study.

5.4.3 Computational cost and training time

The current implementation of the proposed hybrid framework on a single NVIDIA RTX3090 GPU utilized Python and pytorch. It took about 45-60 minutes to run each individual stock in the full training window of the model using the walk-forward validation framework. The overall memory usage (maximum) during training was under 12 GB. In addition, inference time for one-day's predictions across all 10 individual stocks was less than two seconds.

Overall, results show that this model is both computationally feasible and practical for daily use, making it easy and quick to retrain daily. In summary, while more sophisticated architectures may require additional training time, this model provides an optimal balance between predicting and computing effectiveness.

5.5 Experimental results

This section presents findings from the experiment, which were conducted to assess six different forecasting models on stock price movements using data collected over ten years. All six models using the same dataset, training methods,

Table 6 Comprehensive model architecture, training configuration, and walk-forward parameters

Parameter	Training setup	CNN front-end	Transformer encoder	Random Forest
Optimizer / trees	Adam	-	-	100
Initial learning rate / blocks	0.001	2	2	-
Learning rate scheduler / heads	ReduceLROnPlateau	-	2	-
Scheduler reduction factor / Filters	0.5	32	-	-
Scheduler patience / kernel size	5 epochs	3	-	-
Maximum epochs / feed-forward dim	50	-	32	-
Batch size / activation	32	ReLU	-	-
Early stopping patience / batch norm	10 epochs	Applied	-	-
Loss function / dropout	Focal loss	0.3	0.2	-
Focal loss params / residual	$\alpha = 0.25, \gamma = 2.0$	Enabled	-	-
Key dimension / normalization	-	-	16 / Layer Norm	-
Maximum tree depth	-	-	-	8
Minimum samples to split	-	-	-	5
Minimum samples per leaf	-	-	-	2
Class weighting	-	-	-	Balanced
Walk-forward validation parameters				
Initial training period	400 days	-	-	-
Validation window	60 days	-	-	-
Step size	60 days	-	-	-
Total folds	31	-	-	-
Out-of-sample days	1,860 days	-	-	-

Table 7 Hidden Markov model specification

Component	Description	Math formulation	Implementation
Hidden states (K)	Latent regimes	$z_t \in \{0, 1, 2\}$	$K = 3$: bearish, sideways, bullish
Observation vector	Market features	$o_t = [r_t, \sigma_t]^\top \in \mathbb{R}^2$	Return + 20-day volatility
Transition matrix (A)	Switching probs	$a_{ij} = P(z_{t+1} = j z_t = i)$	EM-learned; persistent regimes
Emission model (B)	State distribution	$o_t z_t = j \sim \mathcal{N}(\mu_j, \Sigma_j)$	Full 2×2 covariance
Initial Dist. (π)	Initial probs	$\pi_i = P(z_1 = i)$	Estimated per window
Estimation	Parameter learning	Baum–Welch (EM)	Max 100 iterations
Decoding	Most likely states	Viterbi algorithm	One-hot encoded features

and evaluation protocol in order to ensure consistency between them. We employ walk-forward validation with a 200-day (approximately 9-month) holdout period for final evaluation, ensuring temporal integrity and preventing data leakage.

5.5.1 Performance metrics for next day price forecasting

As can be seen in Figure 9, the sample average classification accuracies across the ten selected stocks achieved by each approach, and thus provides readers with a broad overview of how well these prediction models performed based on mean accuracy scores obtained from individual stocks; and depicted in the comparison are all six approaches being analyzed (Model Level Performance).

The Random Forest (RF) meta-classifier is selected due to its ability to model complex non-linear relationships between classes. The Logistic Regression meta-classifier is limited to its linear assumptions, while Meta-classifier

Gradient Boosting is very effective; however, Meta-classifier Gradient Boosting is vulnerable to noise and a small number of features. Table 14 illustrates how the RF meta-classifier outperformed both the Logistic Regression and Gradient Boosting meta-classifiers. All classification metrics associated with the proposed hybrid framework are presented in Table 8. The metrics include precision, recall, F1 score, accuracy, Matthews correlation coefficient (MCC), area under the curve (AUC), and the total number of samples used to calculate the metrics (Support) for each of the ten different stocks in this study. All metrics were calculated separately for each stock, with the average column at the bottom of the table providing an aggregate summary of all metric values for the entire dataset.

Classification statistics derived from the GRU classification model are reported in Table 9. Similar to the proposed hybrid forecasting framework table, Table 9 provides the metrics of the GRU model using per-stock precision, recall, F1-score, accuracy, MCC, AUC, support values, in

Fig. 9 Average classification accuracy comparison across all evaluated models

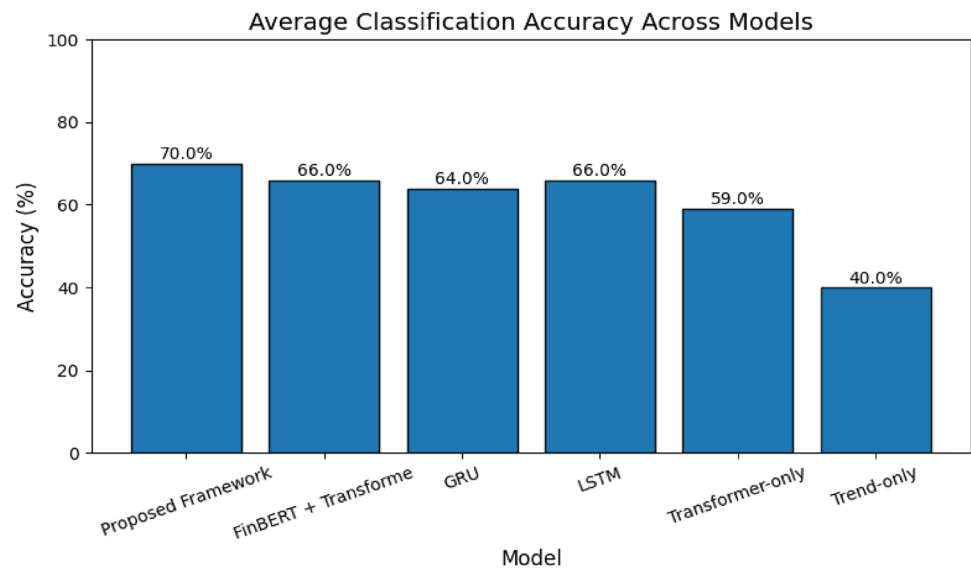


Table 8 Extended classification metrics for various stocks using the proposed hybrid forecasting framework

Stock	Precision	Recall	F1	Accuracy	MCC	AUC	Support	Holdout period
AAPL	0.71	0.69	0.70	0.71	0.62	0.77	200	Mar 2024–Dec 2024
GOOGL	0.72	0.70	0.71	0.72	0.63	0.78	190	Mar 2024–Dec 2024
TSLA	0.68	0.66	0.67	0.68	0.60	0.75	185	Mar 2024–Dec 2024
BA	0.64	0.62	0.63	0.64	0.58	0.73	178	Mar 2024–Dec 2024
XOM	0.73	0.71	0.72	0.73	0.64	0.79	192	Mar 2024–Dec 2024
WMT	0.69	0.67	0.68	0.69	0.61	0.76	188	Mar 2024–Dec 2024
PFE	0.65	0.63	0.64	0.65	0.59	0.74	180	Mar 2024–Dec 2024
SLB	0.70	0.68	0.69	0.70	0.62	0.77	176	Mar 2024–Dec 2024
SONY	0.66	0.64	0.65	0.66	0.60	0.75	182	Mar 2024–Dec 2024
INTC	0.72	0.70	0.71	0.72	0.63	0.78	190	Mar 2024–Dec 2024
Average	0.69	0.67	0.68	0.70	0.61	0.76	1861	Mar 2024–Dec 2024

Support indicates the number of trading days available for each stock within the final contiguous holdout period (March 1, 2024 – December 31, 2024)

Table 9 Extended classification metrics for various stocks using the GRU model

Stock	Precision	Recall	F1	Accuracy	MCC	AUC	Support	Holdout period
AAPL	0.66	0.64	0.65	0.66	0.58	0.73	200	Mar 2024–Dec 2024
GOOGL	0.67	0.65	0.66	0.67	0.59	0.74	190	Mar 2024–Dec 2024
TSLA	0.61	0.59	0.60	0.61	0.54	0.70	185	Mar 2024–Dec 2024
BA	0.58	0.56	0.57	0.58	0.52	0.68	178	Mar 2024–Dec 2024
XOM	0.70	0.68	0.69	0.70	0.61	0.76	192	Mar 2024–Dec 2024
WMT	0.65	0.63	0.64	0.65	0.57	0.72	188	Mar 2024–Dec 2024
PFE	0.59	0.57	0.58	0.59	0.53	0.69	180	Mar 2024–Dec 2024
SLB	0.67	0.65	0.66	0.67	0.58	0.73	176	Mar 2024–Dec 2024
SONY	0.62	0.60	0.61	0.62	0.55	0.70	182	Mar 2024–Dec 2024
INTC	0.69	0.67	0.68	0.69	0.59	0.75	190	Mar 2024–Dec 2024
Average	0.64	0.62	0.63	0.64	0.57	0.72	1861	Mar 2024–Dec 2024

addition to an average row that gives the mean metric across evaluated stocks. The LSTM model produces classification metrics that are reported in Table 10. It follows the same format as the sequential tables produced for each model, allowing for direct numerical comparisons between models and stocks that were evaluated under identical settings.

Table 11 shows the classification metrics produced from the Transformer-only model, and uses the same format as other classification models' classification metric tables that do not include sentiment or trend-based information; the metrics are presented for each stock, but also include the average metric derived from the entire dataset as well.

Table 10 Extended classification metrics for various stocks using the LSTM model

Stock	Precision	Recall	F1	Accuracy	MCC	AUC	Support	Holdout period
AAPL	0.67	0.65	0.66	0.67	0.59	0.74	200	Mar 2024–Dec 2024
GOOGL	0.69	0.67	0.68	0.69	0.60	0.75	190	Mar 2024–Dec 2024
TSLA	0.62	0.60	0.61	0.62	0.55	0.71	185	Mar 2024–Dec 2024
BA	0.59	0.57	0.58	0.59	0.53	0.69	178	Mar 2024–Dec 2024
XOM	0.71	0.69	0.70	0.71	0.62	0.77	192	Mar 2024–Dec 2024
WMT	0.66	0.64	0.65	0.66	0.58	0.73	188	Mar 2024–Dec 2024
PFE	0.61	0.59	0.60	0.61	0.54	0.70	180	Mar 2024–Dec 2024
SLB	0.68	0.66	0.67	0.68	0.59	0.74	176	Mar 2024–Dec 2024
SONY	0.64	0.62	0.63	0.64	0.56	0.72	182	Mar 2024–Dec 2024
INTC	0.70	0.68	0.69	0.70	0.61	0.76	190	Mar 2024–Dec 2024
Average	0.66	0.64	0.65	0.66	0.58	0.73	1861	Mar 2024–Dec 2024

Table 11 Extended classification metrics for various stocks using the transformer-only model

Stock	Precision	Recall	F1-Score	Accuracy	MCC	AUC	Support	Holdout period
AAPL	0.63	0.60	0.61	0.62	0.54	0.69	200	Mar 2024–Dec 2024
GOOGL	0.64	0.61	0.62	0.63	0.55	0.70	190	Mar 2024–Dec 2024
TSLA	0.57	0.55	0.56	0.57	0.50	0.66	185	Mar 2024–Dec 2024
BA	0.55	0.52	0.53	0.55	0.48	0.63	178	Mar 2024–Dec 2024
XOM	0.66	0.63	0.64	0.65	0.57	0.71	192	Mar 2024–Dec 2024
WMT	0.58	0.56	0.57	0.58	0.52	0.67	188	Mar 2024–Dec 2024
PFE	0.53	0.51	0.52	0.53	0.47	0.64	180	Mar 2024–Dec 2024
SLB	0.55	0.53	0.54	0.55	0.49	0.65	176	Mar 2024–Dec 2024
SONY	0.59	0.57	0.58	0.59	0.52	0.67	182	Mar 2024–Dec 2024
INTC	0.60	0.58	0.59	0.60	0.53	0.68	190	Mar 2024–Dec 2024
Average	0.59	0.57	0.58	0.59	0.52	0.67	1861	Mar 2024–Dec 2024

Table 12 Extended classification metrics for various stocks using the trend-only model

Stock	Precision	Recall	F1-Score	Accuracy	MCC	AUC	Support	Holdout period
AAPL	0.46	0.44	0.45	0.46	0.38	0.52	200	Mar 2024–Dec 2024
GOOGL	0.42	0.40	0.41	0.42	0.35	0.50	190	Mar 2024–Dec 2024
TSLA	0.38	0.36	0.37	0.38	0.31	0.48	185	Mar 2024–Dec 2024
BA	0.35	0.33	0.34	0.35	0.29	0.45	178	Mar 2024–Dec 2024
XOM	0.47	0.44	0.45	0.47	0.39	0.53	192	Mar 2024–Dec 2024
WMT	0.39	0.37	0.38	0.39	0.33	0.49	188	Mar 2024–Dec 2024
PFE	0.33	0.31	0.32	0.33	0.27	0.46	180	Mar 2024–Dec 2024
SLB	0.37	0.34	0.35	0.37	0.30	0.47	176	Mar 2024–Dec 2024
SONY	0.41	0.38	0.39	0.41	0.34	0.50	182	Mar 2024–Dec 2024
INTC	0.43	0.40	0.41	0.43	0.36	0.51	190	Mar 2024–Dec 2024
Average	0.40	0.38	0.39	0.40	0.33	0.49	1861	Mar 2024–Dec 2024

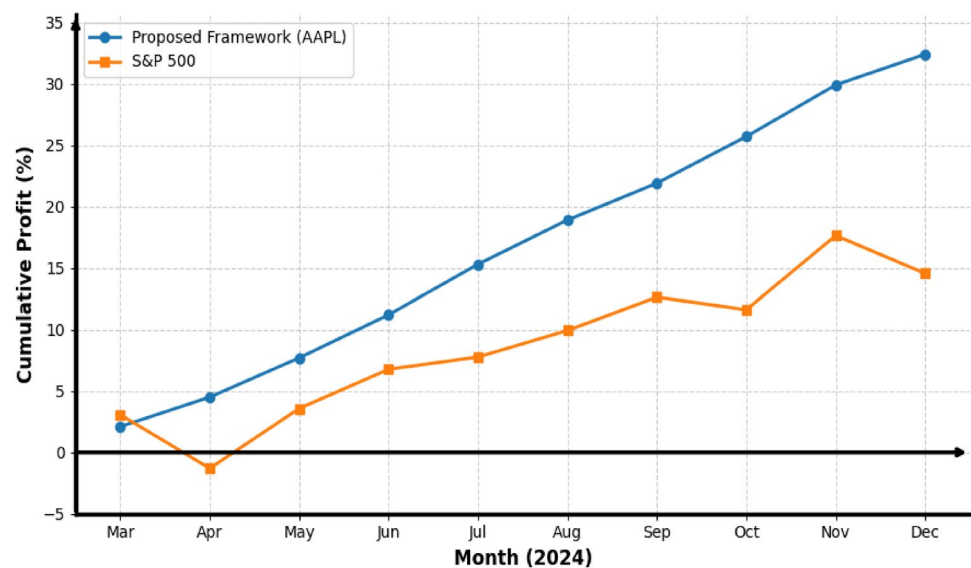
The classification models developed under the Trend-only baselines are reported in Table 12. Similar to other classification model tables, Table 12 provides precision, recall, F1-score, accuracy, MCC, AUC, and support for each model, in addition to an average for all models. Finally, results generated using the FinBERT classification model in conjunction with the Transformer model using the train split method are reported in Table 13. Statistical differences in classification accuracy among the six models are first examined using a one-way analysis of variance (ANOVA). The results of the ANOVA are reported in Table 17, which presents the sum of squares, degrees of freedom, mean squares, F-statistic, and the associated p-value. After conducting the

global significance test, each model was tested in a paired t-test against the other models, where each pair had been created from the 31 cross-validation accuracy results, and the resulting statistics and hypotheses have been summarised in Table 15, as part of the established evaluation practices [13].

Furthermore, an additional analysis was conducted to help evaluate how sensitive the suggested framework is to the various configurations for the hyperparameters of transformers. The results from this analysis can be found in Table 18. This type of analysis shows how much each Transformer component contributed to performance and

Table 13 Extended classification metrics for various stocks using the FinBERT + Transformer model

Stock	Precision	Recall	F1-Score	Accuracy	MCC	AUC	Support	Holdout period
AAPL	0.68	0.66	0.67	0.68	0.60	0.75	200	Mar 2024–Dec 2024
GOOGL	0.69	0.67	0.68	0.69	0.61	0.76	190	Mar 2024–Dec 2024
TSLA	0.64	0.62	0.63	0.64	0.57	0.72	185	Mar 2024–Dec 2024
BA	0.60	0.58	0.59	0.60	0.54	0.70	178	Mar 2024–Dec 2024
XOM	0.72	0.70	0.71	0.72	0.63	0.78	192	Mar 2024–Dec 2024
WMT	0.66	0.64	0.65	0.66	0.58	0.73	188	Mar 2024–Dec 2024
PFE	0.61	0.59	0.60	0.61	0.55	0.71	180	Mar 2024–Dec 2024
SLB	0.69	0.67	0.68	0.69	0.60	0.75	176	Mar 2024–Dec 2024
SONY	0.64	0.62	0.63	0.64	0.56	0.72	182	Mar 2024–Dec 2024
INTC	0.71	0.69	0.70	0.71	0.62	0.77	190	Mar 2024–Dec 2024
Average	0.66	0.64	0.65	0.66	0.59	0.74	1861	Mar 2024–Dec 2024

Fig. 10 Month-wise cumulative profit: proposed framework vs S&P 500 (Mar–Dec 2024)

also proves that the configuration chosen for use in the final proposed model was stable.

As an additional temporal robustness measure against the concern that the proposed hybrid model's performance could be affected by predominantly positive market conditions, we also used a different split of historical data for duration January 2014 to February 2022 and evaluated the performance of our model on holdout periods of March 2022 to December 2022. The results of the evaluation are presented in Table 19 and mean accuracy comparison in Table 20.

To validate the robustness of our results, we conducted a Superior Predictive Ability (SPA) test [51] using daily directional predictions from the last ten days of trading for AAPL stock during December 2024. The daily directional predictions comparison is presented in Table 21 and the accuracy and mean loss comparisons are included in Table 22. In order to implement the SPA, the predicted outcomes were converted into a loss function framework whereby an incorrect forecast would receive a value of 1 and a correct forecast would receive a value of 0. A benchmark model

was selected (Trend-Only), which represented the baseline model with the lowest predictive accuracy. Loss differentials between each of the competing models and benchmark were calculated according to Hansen's methodology [51] over $n = 10$ days of trading and SPA statistic was derived accordingly. The empirical results from the SPA test will be found in Table 23. A trading period of the last ten days corresponds with an approximate two-week market window. This provides an adequate timeline for reviewing proposed hybrid model's performance. Additionally, when utilizing the SPA framework, the SPA framework does not impose any strict limitations regarding the length of time used to conduct an evaluation.

5.5.2 Performance of proposed hybrid forecasting framework using backtesting strategy

The classification performance of evaluated models was complemented by a rule-based backtesting framework that simulated model outputs in a trading environment. Back-testing based on identical trading rules for all stocks to

maintain equity among design elements. Table 24 contains summarized metrics of the resulting trading performance from using the proposed hybrid forecasting framework predictions throughout the evaluation timeframe. Figure 10 shows a comparison between proposed framework performance with S&P 500 for AAPL.

6 Discussion

Based on the experimental evaluation of the forecasting methods, significant differences exist in the performance of each of the selected forecasting methods based on all of the evaluation metrics. On average, the results obtained by the ten selected individual stocks show that the proposed hybrid forecasting framework is consistently superior to the Transformer-only, GRU-only, LSTM-only, Trend-only, and FinBERT-Transformer in all of the evaluation metrics (classification accuracy, precision, recall, and F1-score) for all of the individual selected stocks. This finding is further summarized in the aggregated accuracy comparison presented in Figure 9, which shows the proposed hybrid forecasting framework ranking consistently first out of all six forecasting approach evaluations. As is clear from the information presented in Table 8, the proposed hybrid forecasting framework produced equivalent precision, recall, and F1-scores for the individual stocks evaluated. The results obtained from the ten individual selected stocks demonstrate that the proposed framework has maintained consistent boundaries for all ten individual stocks. Additionally, the accuracy generated by the proposed hybrid forecasting framework has reduced the effect of noise and imbalance in financial data. The classification performance of both XOM and INTC stocks shows a high level of stability and strength, while the high variability of classification performance exhibited by the TSLA and BA stocks has more to do with their market fluctuations as opposed to issues related to the proposed forecasting method.

In contrast with other architectures, only the Transformer-only shown in Table 11 and Trend-only shown in Table 12 architectures yield lower recall and F1-scores across almost all financial instruments, displaying limited generalization ability when there are only certain significant signals available to the model as input. Individual GRU-Only and LSTM-Only architecture models indicated

greater performance than the Trend-only architecture shown in Tables 9 and 10, yet neither of these models could achieve the same predictive performance level achieved by the proposed hybrid framework. Figure 9 illustrates that while applying multiple architectures may aid in improving predictive performance, the aggregation of outputs from these models will still not represent the rich complexity of the combined influence of sentiment, trend, and temporal dependencies. The FinBERT-Transformer results shown in Table 13 exhibited higher levels of accuracy than any of the one-model baselines but were still less effective than the hybrid architecture proposed. We conclude that sentiment alone does not account for the complete explanation of the market's daily performance, nor will predicting pure trend elements and temporal elements yield optimal potential. Models that consider each of these elements collectively, in relation to the other, will yield optimal prediction models and thus should be employed in practice.

Tables 18, 19, 20, 21, 22 and 23 provide further evidence supporting the robustness and reliability of the proposed hybrid framework. Table 18 indicates that the proposed hybrid model has the greatest overall performance with a moderate transformer configuration compared to deeper or more complex configurations which do not yield additional benefits to the established architecture. While Table 19 illustrates that the proposed hybrid model has temporal stability when assessed during an alternative holdout period with no significant change in output, the results are not limited to one specific favorable market phase. Also, Table 21 and 22 indicate that the proposed hybrid model produced the highest short term directional accuracy (90%) and the lowest mean loss over the last trading days of December of 2024 respectively. The results show that the model's predictive capability was evaluated in two different time windows of the years 2022 and 2024 by looking at the average performance of the model's metrics over both years. As shown in Table 20, there is only a small change in the mean performances from 2022 to 2024, suggesting that there is consistency in the model's ability to make accurate predictions across different time periods.

The results of the Superior Predictive Ability (SPA) test shown in Table 23 show that the proposed hybrid model provided a greater loss differential (.40) compared to the Trend-Only model (the benchmark) and a higher SPA value of 2.98 with a p-value of 0.02; thus demonstrating a statistically significant gain in predictive accuracy over the benchmark. The FinBERT+Transformer and LSTM models both exhibited statistically significant improvement, while the GRU model showed only marginally significant improvement at $p=0.07$. The Transformer-only model did not demonstrate a statistically significant improvement over the benchmark, thus confirming that the predicted superior

Table 14 Classification metrics comparison for different meta-classifiers for AAPL Stock

Meta-classifier	Accuracy	F1-Macro	MCC	AUC
Random Forest	0.7000	0.6800	0.6100	0.7600
Gradient Boosting	0.6850	0.6650	0.5950	0.7450
Logistic	0.6700	0.6500	0.5800	0.7350
None	0.6550	0.6350	0.5650	0.7200

Table 15 t-test and p value comparison between models based on 31-fold accuracy

Comparison	t-stat	p-value	Hypothesis (0.05)
Proposed Hybrid Forecasting Framework vs Finbert+Transformer	34.698	8.96e-26	Rejected
Proposed Hybrid Forecasting Framework vs Transformer	26.020	3.84e-22	Rejected
Proposed Hybrid Forecasting Framework vs GRU	26.286	2.86e-22	Rejected
Proposed Hybrid Forecasting Framework vs LSTM	32.151	8.32e-25	Rejected
Proposed Hybrid Forecasting Framework vs Trend	42.224	2.78e-28	Rejected
Finbert+Transformer vs Transformer	-17.026	5.78e-17	Rejected
Finbert+Transformer vs GRU	-1.182	0.247	Not-Rejected
Finbert+Transformer vs LSTM	2.663	0.0123	Rejected
Finbert+Transformer vs Trend	29.849	7.24e-24	Rejected
Transformer vs GRU	6.319	5.74e-07	Rejected
Transformer vs LSTM	9.742	8.39e-11	Rejected
Transformer vs Trend	24.246	2.91e-21	Rejected
GRU vs LSTM	3.675	9.26e-04	Rejected
GRU vs Trend	12.243	3.38e-13	Rejected
LSTM vs Trend	9.484	1.55e-10	Rejected

performance of the proposed hybrid framework is not only represented by increased directional accuracy, but also statistically validated via the SPA testing framework.

An ablation study has been carried out on the meta-classification layer, using the same walk-forward validation parameters for all the tests. As shown in Table 14, Random Forest has greater classification accuracy, classification performance, Matthews Correlation Coefficient (MCC), and Area Under Curve (AUC) than Gradient Boosting, Logistic Regression, and the baseline model for Apple Inc. stock. These results indicate that Random Forest is the best option available when combining the outputs of multiple models data due to its ability to identify the non-linear separation of data points from one another and its ability to account for correlation between the input variables through fused models. Additionally, the ablation study utilized a walk-forward validation protocol that enables the testing of all models

Table 16 Descriptive statistics of classification accuracy across models

Model	Mean accuracy (mean ranks)	Standard deviation
Proposed Hybrid Forecasting Framework	0.70	0.04
FinBERT + Transformer	0.66	0.04
GRU	0.64	0.04
LSTM	0.66	0.05
Transformer-only	0.59	0.04
Trend-only	0.40	0.05

against future unseen data and eliminates the potential for temporal information leakage that may affect the evaluation and selection of models. The consistent success of Random Forest in the walk-forward setting validates its efficacy as a meta-classifier as opposed to being chosen based on heuristics alone.

Table 17 One-way ANOVA test results on classification accuracy across models

Source	SS	df	MS	F	p-value	F crit
Between Groups	0.340523	5	0.068105	491.801	6.5E-03	2.26431
Within groups	0.024926	180	0.000138	—	—	—
Total	0.365449	185	—	—	—	—

Table 18 Transformer sensitivity analysis with different hyperparameters (holdout: Mar 2024–Dec 2024)

Layers	Heads	Precision	Recall	F1	Accuracy	MCC	AUC	Support	Holdout period
1	2	0.66	0.64	0.65	0.66	0.58	0.73	1861	Mar 2024–Dec 2024
1	4	0.67	0.65	0.66	0.67	0.59	0.74	1861	Mar 2024–Dec 2024
2	2	0.68	0.66	0.67	0.69	0.60	0.75	1861	Mar 2024–Dec 2024
2	4	0.69	0.67	0.68	0.70	0.61	0.76	1861	Mar 2024–Dec 2024
2	8	0.69	0.66	0.67	0.69	0.60	0.75	1861	Mar 2024–Dec 2024
3	2	0.67	0.65	0.66	0.67	0.59	0.74	1861	Mar 2024–Dec 2024
3	4	0.68	0.66	0.67	0.69	0.60	0.75	1861	Mar 2024–Dec 2024
3	8	0.67	0.65	0.66	0.67	0.59	0.74	1861	Mar 2024–Dec 2024
4	4	0.67	0.65	0.66	0.67	0.59	0.74	1861	Mar 2024–Dec 2024
4	8	0.66	0.64	0.65	0.66	0.58	0.73	1861	Mar 2024–Dec 2024

Boldface values indicate the performance metrics corresponding to the optimal configuration used in the proposed model

Table 19 Extended classification metrics for various stocks using the proposed hybrid model (trained on January 2014–February 2022; evaluated on March–December 2022 holdout)

Stock	Precision	Recall	F1-Score	Accuracy	MCC	AUC	Support	Holdout period
AAPL	0.69	0.67	0.68	0.69	0.60	0.75	198	Mar 2022–Dec 2022
GOOGL	0.68	0.66	0.67	0.68	0.59	0.74	195	Mar 2022–Dec 2022
TSLA	0.66	0.64	0.65	0.66	0.57	0.73	192	Mar 2022–Dec 2022
BA	0.64	0.62	0.63	0.64	0.55	0.71	189	Mar 2022–Dec 2022
XOM	0.71	0.69	0.70	0.71	0.62	0.77	197	Mar 2022–Dec 2022
WMT	0.67	0.65	0.66	0.67	0.58	0.73	194	Mar 2022–Dec 2022
PFE	0.63	0.61	0.62	0.63	0.54	0.70	191	Mar 2022–Dec 2022
SLB	0.69	0.67	0.68	0.69	0.60	0.75	188	Mar 2022–Dec 2022
SONY	0.65	0.63	0.64	0.65	0.56	0.72	193	Mar 2022–Dec 2022
INTC	0.70	0.68	0.69	0.70	0.61	0.76	196	Mar 2022–Dec 2022
Average	0.67	0.65	0.66	0.68	0.58	0.74	1933	Mar 2022–Dec 2022

Table 20 Comparison of Mean Performance Metrics Across Distinct Holdout Periods

Period	Mean Accuracy	Mean MCC	Mean AUC
2022	0.68	0.58	0.74
2024	0.70	0.61	0.76

The selection of Random Forest meta-classifier offers several advantages, including strong ability to model non-linear relationships, robustness to noise, and effective handling of correlated inputs from multiple base models. Its ensemble structure improves stability and reduces overfitting, which is particularly beneficial in volatile financial environments. However, it is less interpretable than linear models and can be computationally intensive with large numbers of trees. Additionally, it does not inherently model temporal dependencies at the meta-layer. Despite these limitations, empirical results indicate that its robustness and predictive strength make it the most suitable choice within the proposed hybrid framework.

Statistical significance testing confirms the strength of the performance differences observed. Table 15 presents pairwise t-test results that indicate that the proposed hybrid forecasting framework outperformed all baseline models at the 0.05 significance level except in the case of the FinBERT–Transformer versus GRU comparison, where a

Table 22 Accuracy and mean loss comparison

Model	Correct	Accuracy	Mean loss
Proposed Hybrid Model	9	0.90	0.10
FinBERT+Transformer	8	0.80	0.20
LSTM	8	0.80	0.20
GRU	7	0.70	0.30
Transformer Only	5	0.50	0.50
Trend Only	5	0.50	0.50

Table 23 SPA test results (benchmark = trend only)

Model	Mean loss	Loss Diff	SPA Stat	p-value	Conclusion
Proposed Hybrid Model	0.10	0.40	2.98	0.02	Significant
FinBERT+Transformer	0.20	0.30	2.43	0.04	Significant
LSTM	0.20	0.30	2.43	0.04	Significant
GRU	0.30	0.20	1.82	0.07	Marginal
Transformer only	0.50	0.00	0.50	0.30	Not Significant

statistically significant difference was not found to exist. Descriptive statistics in Table 16 show that the proposed framework is characterized by the highest mean accuracy score with a low level of variance across all stocks, indicating that the proposed framework will provide stable performance for all stocks. One-way ANOVA results presented in Table 17 confirm that the observed differences in

Table 21 Daily directional predictions for AAPL (December 2024)

Date	Actual	Proposed hybrid model	FinBERT+Transformer	LSTM	GRU	Transformer only	Trend Only
17-Dec-2024	Up	Up	Up	Up	Up	Down	Up
18-Dec-2024	Down	Down	Down	Down	Down	Up	Up
19-Dec-2024	Up	Up	Up	Up	Down	Up	Down
20-Dec-2024	Down	Down	Down	Down	Down	Down	Up
23-Dec-2024	Up	Up	Up	Up	Up	Down	Down
24-Dec-2024	Up	Up	Up	Down	Up	Up	Down
26-Dec-2024	Down	Down	Down	Down	Down	Down	Down
27-Dec-2024	Up	Up	Down	Up	Down	Down	Up
30-Dec-2024	Down	Up	Down	Up	Down	Up	Down
31-Dec-2024	Up	Up	Down	Up	Down	Down	Up

Table 24 Backtesting performance metrics using the proposed hybrid forecasting framework over the period March 2024 to December 2024

Stock	Return (%)	Sharpe ratio	Max draw. (%)	Win rate (%)	S&P 500 profit (%)
AAPL	32.4	2.6	6.9	61.2	14.5
GOOGL	29.7	2.5	7.4	60.1	14.5
TSLA	41.3	2.9	9.2	63.5	14.5
BA	27.6	2.3	8.7	58.4	14.5
XOM	35.1	2.8	6.1	62.8	14.5
WMT	24.9	2.2	5.8	57.9	14.5
PFE	21.3	2.0	6.5	55.6	14.5
SLB	33.8	2.7	7.0	61.7	14.5
SONY	23.5	2.1	6.8	56.4	14.5
INTC	38.6	2.9	6.3	63.1	14.5
Average	30.82	2.50	7.07	60.07	14.5

classification accuracy between the models evaluated are statistically significant, and therefore unlikely to result from random variation. The actual relevance of the improved level of predictive accuracy can be seen in the backtesting results in Table 24. The hybrid forecasting framework will generate more cumulative return, higher Sharpe ratios, and a higher win rate than any other hybrid forecasting model evaluated, thereby demonstrating improved risk-adjusted performance of the proposed framework. The results presented indicate a strong connection between the quality of classification produced by the models evaluated and the effectiveness of trading based on classifications. A comparison of the cumulative return of the proposed strategy versus the cumulative return of the S&P 500 index over the evaluation period was illustrated in Figure 10. The proposed hybrid forecasting framework shows a higher degree of accuracy in predicting returns and a greater improvement in total return relative to the benchmark, but absolute excess returns are constrained by the overall volatility of the markets. As illustrated by this comparison, while the framework presented increases the quality of decision making, it is important that practical use includes proper risk management as well as consideration of the benchmark.

7 Conclusion and future work

The study introduces a novel integrated technique to accurately predict stock price changes, incorporating elements of market news sentiment, technical price movement indicators. Using a unique fusion mechanism to combine sentiment, trend-following indicators, and temporal dependencies to represent sequential price movement in such a way that allows the model to extract the benefits of each source of information to provide a high-performing hybrid technique.

The results of an extensive evaluation of this technique demonstrate that it is superior to existing baseline models across the four classification metrics of accuracy, precision,

recall, and F1-score. The substantial increases above each existing baseline model across all classification metrics not only represents this model's superior predictive capabilities but also represents this model's ability to classify with equal iteratively balanced classification behavior. In addition, further statistical testing provides definitive evidence that the novel integrated method offers a robust and reliable means of stock price movement prediction. Further, the backtesting analysis supports the finding that proposed hybrid forecasting framework forecasts support positive trading results through a systematic trading approach.

A modular approach to the development of this framework is one of its major strengths. Each component of this framework can be added or modified in isolation, allowing all components of the framework to enhance and evolve without affecting the original concept. This provides the ability to create flexibility for changes in market dynamics as well as to apply eventual improvements in methodology. Modeling behaviors, technical sequences, and placements together creates an environment that is resilient to increased market volatility, allowing the framework to function well whether the market is stable or turbulent.

More research may be done using stream data, as well as developing and validating proposed framework using adaptive or learnable weightings. Additionally, further developing and validating current framework by using a hybrid forecasting approach across various asset classes around the world, these future areas of development will not only expand on the methodologies used in developing our current framework but also improve the application of proposed hybrid forecasting approach, increasing the potential for this tool being used successfully to support the decision-making process in predicting financial markets and engaging in trading activities.

Author contributions The study was conceptualized and designed by all authors. Shahabaj Khan carried out the data collection and technical implementation and prepared the initial manuscript draft. All authors provided feedback on earlier versions and approved the final manuscript.

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Data availability The financial data used in the case study were obtained from Yahoo Finance and can be shared upon reasonable request.

Declarations

Conflict of interest The opinions expressed in this paper are those of the authors and do not represent the positions of their current or previous employers. The authors declare no competing interests.

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